Specification Surfaces for Incremental forecast performance: Estimation, Uncertainty, and Multi-Log Evaluation

Abstract

What a recorded field is worth to a prediction is reported as one number, but it is a functional of choices the report rarely states: the baseline, the pipeline, the forecast target, the split, the moment the decision is taken, how completely the register behind the field is populated, and the metric. We define incremental forecast performance as a *specification surface* $V_s(f)$ over a design space the analyst declares, and give three objects for reporting one: an exact functional-ANOVA decomposition of its variance that separates analyst choices from resampling; simultaneous max-t bands from a pipeline-refitting bootstrap, at a critical value with measured coverage; and *resolution regions*, which say where a sign is determined. On 19 pairs from 13 public event logs, resampling carries a median 71.7% of the variance, and net of it which analyst choice leads is a property of the pair. Most specifications leave the sign undetermined; where the data determine one, the resolved specifications sit a median 0.0600 from the conventional report in area under the receiver operating characteristic curve (AUC). In a case study on 45,455 bank incidents, a register's worth turns on the forecast origin and its sign on the target.

Keywords: specification surface, incremental forecast performance, event logs, simultaneous inference, sensitivity analysis, configuration management database

Highlights

- A field's incremental predictive performance is a surface, not one number
- An exact decomposition separates analyst choices from resampling

- Simultaneous bands over the whole declared family, with coverage measured
- Resolved cells sit 0.0600 in area under the curve from the one-number report
- A register's worth turns on the decision time and on which target is set

1. Introduction

A claim that a recorded field is worth something is a comparison. Its second arm — performance *without* the field — is a choice the analyst makes, and so are the learner that makes the comparison, the encoding it uses, the outcome it predicts, the way the data are split in time, whether the field is available at the moment the decision is taken, how completely the register behind the field is populated, the metric the difference is read in, and the operating point at which that metric is evaluated. Each is routinely left implicit, and the result is reported as one number.

This paper's position is not that such a number depends on those choices — that is known, and Section 2 says by whom — but that the dependence is large enough, on real decisions, that the number alone is not a report; that the right object is therefore a surface over a *declared* design space; and that a surface admits estimation and inference of its own. Two things follow that a reader may not expect, and both are stated here before anything is argued for. **A pooled specification surface overstates the case**: most of its variance here is resampling rather than analyst latitude, and the correction is exact (Section 4.5). And **the sign is the wrong thing to count**: where the data determine a sign, the conventional report is usually right about it and usually silent about the size, which is what Section 6.4 reports instead.

The estimand. Write a specification s for a complete assignment to every design axis, and

$$V_s(f) \;=\; m_\theta(A_s(D_{\text{train}}, B \cup \{f_q\}),\, D_{\text{test}}) \;-\; m_\theta(A_s(D_{\text{train}}, B),\, D_{\text{test}})\,, \quad (1)$$

where A_s is the fitting procedure (learner, encoding, hyperparameters, calibration), B the baseline feature set, f_q the field under a register quality mechanism q, m the metric and θ the operating point, and where $D_{\text{train}}, D_{\text{test}}$ are fixed by the split rule and the cohort. The *specification surface* is the

map $s \mapsto V_s(f)$ over a design space $\mathcal{S}$ that the analyst declares and the reader can enumerate; Figure 1 draws one.

We say *incremental forecast performance* and not *value*: (1) measures how much better a model predicts, not what the field costs to acquire or maintain, whether acting on the prediction changes the outcome, or what an organisation would pay for either. Section 8 converts the increment into net benefit at a declared exchange rate, which is as close to economic value as this evidence gets.

Contributions.

1. **An estimand, a declared surface and an exact decomposition** (Table 1). The surface over a design space the reader can enumerate; three *denominators* distinguished and audited from the files, so that a claim's quantifier is a counted set rather than a phrase; and an exact functional-ANOVA decomposition of the surface's variance in which the split is an *error stratum*, so that resampling is separated from the choices an analyst makes (Section 4.5).
2. **Simultaneous inference whose coverage is measured, and the labels that rest on it.** Simultaneous max-t bands over the whole family a claim ranges over, from a pipeline-refitting bootstrap, at a critical value whose family-wise coverage is measured against a known answer in Section 9.2 — 94.7% on the family matched to this design, under an ideal bootstrap that bounds the critical value's contribution and not the resampling's — and *resolution regions* with a robustness index ρ under a declared minimum resolved share.
3. **A registered multi-log benchmark and a register at three forecast origins.** The surface is computed through one frozen pipeline on every log–target pair a registered protocol admits — 19 of them, not only the ones where the answer resolves — and every corpus statistic is reported on all 19 and on the 8 where the register is a maintained one (Supplement S14.17). Section 7 then shows that a forecast origin fixes which cases are in scope, what is known about them and which fields are admissible at once, reconciles the two values the case study's log has been reported to give, and finds the difference is the *target*: over the cohort × target design the target carries 99.3% of the variance and the cohort 0.0%.

Figure 2 sets the method out once, from the three roles a log is read into to the three denominators a claim may quantify over.

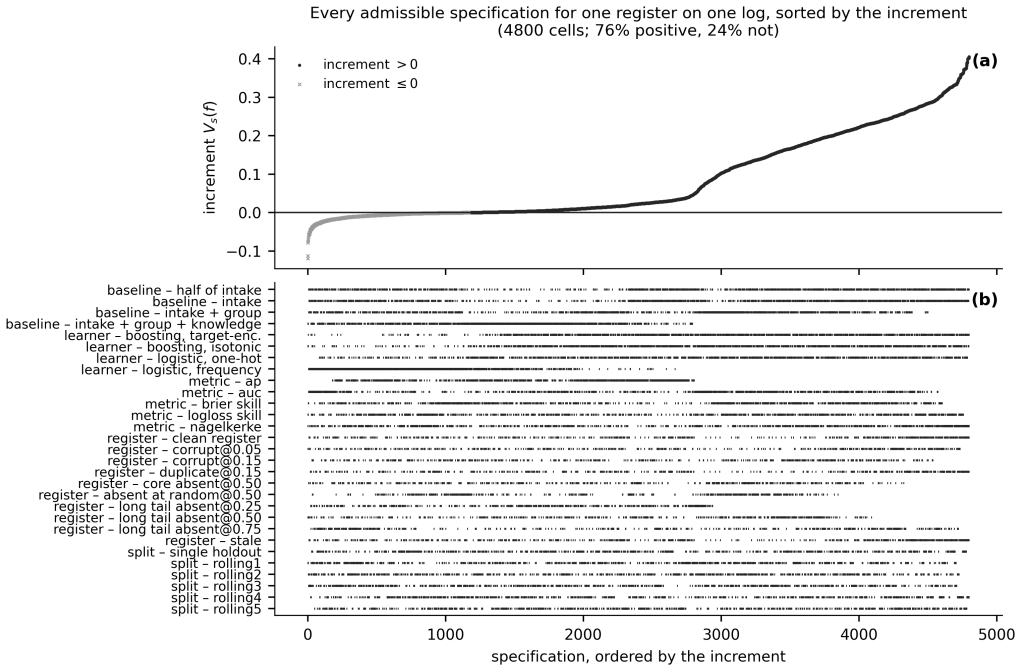


Figure 1: The specification curve for the configuration item on BPI Challenge 2014 (BPIC14), at the registered handover target: 4,800 admissible scalar cells, 4 pipelines × 4 baseline rungs × 5 instruments × 10 register-quality conditions × 6 splits. **(a)** Each point is one admissible specification — a complete assignment to those five axes — sorted by the increment. **(b)** Which level of which axis each specification occupies. The intercept-only rung is excluded. Every point is a point estimate; the simultaneous bands that decide the region labels are in Table S9. **The five scalar instruments are plotted on one axis and their increments are not on a common scale** (Remark 1); what the panel supports is the *sign* statement, which is unit-free, and not a comparison of magnitudes across instruments. The panel shows the increment rather than the reduction, so a cell where the register is worth almost nothing looks the same as one where the free field absorbs almost everything.

Specification regret — what choosing one specification costs against the best, on a scale a decision can use — is reported in full and is *not* among the contributions above. Sections 6.6 and 8.3 report that the one-number rule's median excess regret is zero in every instrument, that the out-of-sample ordering reverses under rolling folds, and that in net benefit the rules separate on most pairs with the excess falling on the conservative rule rather than on the one-number rule. It is carried as a *robustness check* on the other two arguments (Section 4.7): an object that reports a null is a check and not a

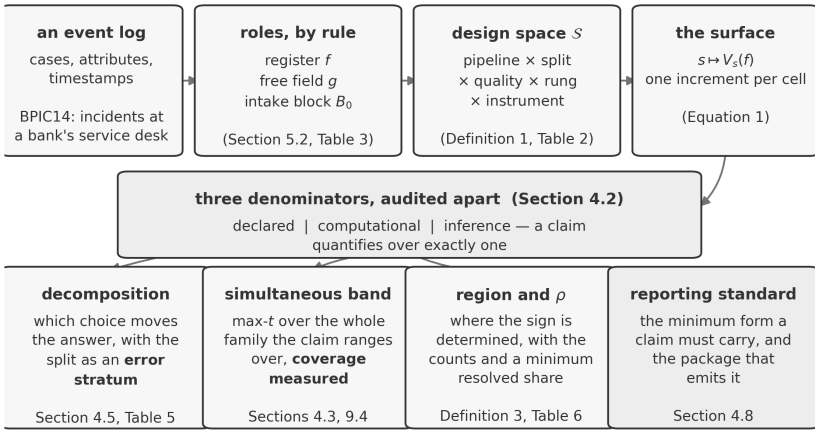


Figure 2: The method, once. A log is read into three *roles* by declared rules; the roles are crossed over a declared *design space*; the result is a *surface*, one increment per cell. Every claim about that surface names which of three *denominators* it quantifies over, and the reporting objects are computed from it. The schematic carries no result: the numbers are in the tables it names.

result.

What this paper is not. This is a retrospective study on public benchmarks, offered as an evaluation methodology. It is not a procurement conclusion about configuration management databases, not a systematic review of reporting practice, and not a report of a deployed system; Section 10 says what each of those would have required.

The supplementary document. The protocol and its amendments, the proofs, the constructions of the reporting objects, the correction register, the reporting standard, the results this article summarises and the tables it does not print are in a supplementary document whose sections, tables and figures carry an S. The article states every claim it makes and prints the evidence for each; a reader who wants to check an inference rather than read one will want the supplement, and Supplement S10 is a glossary of every term used here in a technical sense, with the case study's own value for each.

object	what it is for	§	table	what it costs
Sensitivity decomposition	how much of the disagreement each analysis choice explains, and how much belongs to no single one	4.5	S8	exact; no refit. A Möbius sum over the surface already computed
Simultaneous band	an interval that holds over every cell a claim quantifies over, not one cell at a time	4.3	S9	the whole cost: one draw refits every arm of every cell
Resolution region and ρ	which way the surface points where the data settle it, and how much of it they settle	4.6	6	a function of the band; no further refit
Specification regret *(a check, not an object)*	what choosing one specification costs against the best one, on a scale a decision can use — it returns a null on this corpus	4.7	S27	exact; no refit, on the surface already computed

Table 1: **The reporting objects.** The three this paper supplies and the fourth it retains as a robustness check, what each is for, where it is defined, where it is reported on this corpus, and what it costs; the two middle columns are the section it is defined in and the table that reports it. Only the second costs model fits: a draw of the bootstrap refits every arm of every cell, which is why the surface an inference is taken over is smaller than the surface a point estimate is taken over (Section 4.2). The other three are functions of a surface that has already been computed, so an analyst who has done the sweep has already paid for them.

2. Related work

Every component used here is established. That a feature's apparent importance depends on what else is in the model motivates conditional and permutation importance (Strobl et al., 2008; Fisher et al., 2019), Shapley attributions (Lundberg and Lee, 2017; Covert et al., 2020) and nonparametric formulations of importance as a population quantity (Williamson et al., 2021); the clinical-prediction literature has said for two decades that the increment in the area under the ROC curve (AUC) from adding a predictor depends on the existing model (Pencina et al., 2008; Cook, 2007), is a poor guide to clinical usefulness (Cook, 2007; Vickers, 2008) and is routinely

misread (Kerr et al., 2011); the same argument is made about effort-aware metrics in defect prediction (Çarka et al., 2022), where which metric is chosen changes which method wins. Decision-analytic evaluation supplies the exchange rate (Vickers and Elkin, 2006; Vickers et al., 2016, 2019). Multiverse and specification-curve analysis supply the sweep (Steegen et al., 2016; Simonsohn et al., 2020; Patel et al., 2015), underspecification and variance studies the observation that pipelines differ (D'Amour et al., 2022; Bouthillier et al., 2021), and data valuation a different question about a datum's worth (Ghorbani and Zou, 2019; Jia et al., 2019). On event logs, outcome-oriented predictive monitoring supplies the task (Teinemaa et al., 2019; Verenich et al., 2019; Senderovich et al., 2018), encoding studies one axis (Liu et al., 2025), and data-quality work the vocabulary for a degraded register (Suriadi et al., 2017; Mohammed et al., 2025; Comuzzi et al., 2025). Leakage discipline is standard (Kaufman et al., 2012; Kapoor and Narayanan, 2023; Weytjens and De Weerdt, 2022).

Four further strands bear on the argument and are set out in Supplement S1.1 with what this paper takes from each: *many-analysts studies* (Silberzahn et al., 2018; Breznau et al., 2022), which establish that analytic variability is large without supplying an estimand for it; *predictive multiplicity* and Rashomon sets (Marx et al., 2020; Semenova et al., 2022), which are multiplicity at the level of a fitted model where this paper's is at the level of the design that produced it; *benchmark-variance work in predictive process monitoring* (Rama-Maneiro et al., 2023), which is the evidence this paper's decomposition partitions; and the *sensitivity-analysis* tradition of bounding an unmeasured assumption (VanderWeele and Ding, 2017), which is the form Section 7.3's leakage tipping point takes.

The novelty gap, precisely. Every component used here is established and none is claimed. What none of them supplies is the object this paper reports, and the gap is in four places: the estimand is not declared completely, the forecast origin and the register's quality being treated as preliminaries rather than arguments; the measure is left implicit, a specification curve being summarised uniformly over whatever cells were enumerated, which moves the answer (Section 4.5); inference is pointwise or absent, a curve being a distribution of point estimates and a permutation test on its median, which licenses no statement about a sub-region; and nothing measures what the conventional report costs. What is new is correspondingly narrower than the objects' names suggest. The decomposition adapts functional ANOVA over a

configuration space (Hutter et al., 2014); counting the share of specifications with an effect in a declared direction is standard specification-curve practice (Simonsohn et al., 2020), and the robustness index ρ is that count. What is new is what the count is taken over and under — a *simultaneous* band over a declared family rather than a pointwise test, an *audited* denominator rather than the cells that happened to be enumerated, and a minimum resolved share below which no direction is reported — together with the decomposition that treats the split as an error stratum and so separates analyst latitude from resampling exactly, which we have not seen elsewhere. Section 6.5 measures where a curve's verdict and a region label part company rather than asserting a difference. The contribution is deliberately *not* "people usually report one number"; it is that a claim of this form is a functional of a design, that the functional can be estimated and its uncertainty quantified, and that on this corpus it is far from constant.

3. The specification surface

3.1. The design space

Definition 1 (Design space)**.** *A* design space $\mathcal{S}$ *for a feature-value claim is a finite set of specifications, each a complete assignment to the axes*

$$s = \Big(\underbrace{A}_{\text{learner}}, \underbrace{E}_{\text{encoding}}, \underbrace{Y}_{\text{target}}, \underbrace{\Pi}_{\text{split}}, \underbrace{\tau}_{\text{forecast origin}}, \underbrace{q}_{\text{register quality}}, \underbrace{B}_{\text{baseline}}, \underbrace{m}_{\text{metric}}, \underbrace{\theta}_{\text{operating point}} \Big),$$

together with the cohort and eligibility rule that fix D. A claim about a feature is well posed *only relative to a declared $\mathcal{S}$.*

Declaring a design space means writing down every axis, its levels, the measure over those levels, and every level deliberately excluded with the reason. Table 2 is that declaration, generated from the code that walks the space, and its caption says which rows multiply into the admissible cell counts of Section 6 — the learner, the split, the register-quality condition, the admissible baseline and the scalar instrument — and where each of the others is counted instead; that multiplication, re-done pair by pair from the same file the table is generated from, is a condition the verification harness checks. Not all nine axes are varied in this study: Table 3 states per pair which ran

axis	levels	in the product	measure	exclusions
learner	2–4	yes	equal / ref. / conc.	none
encoding	2–3	no: in the learner	set by the learner	none
split	6	yes	equal / ref. / conc.	none
forecast origin (availability)	1–2	no: second surface	never averaged	two times on the primary log, one elsewhere
register quality by severity	6–10	yes	equal / ref. / conc.	none
baseline (ladder rung)	4–5	no: the row below	equal / ref. / conc.	intercept-only: in the surface, out of every admissible set
baseline, admissible only	3–4	yes	equal / ref. / conc.	none
scalar instrument	5	yes	never averaged across	none
operating point (net benefit)	31	no: decision curve	declared threshold dist.	outside the declared range

Table 2: The declared design space. One row per axis: the number of levels (a range where it differs by log), whether the axis is a **factor of the admissible cell count**, the measure the paper places on those levels, and every level excluded by declaration with its reason. Generated from the code that walks the space. **Not every row multiplies**: only the rows marked 'yes' do, and their product is learner × split × register quality × admissible baseline × scalar instrument, which is the 'learners', 'splits', 'quality', 'rungs' and 'metrics' columns of Table S14 and equals its 'declared scalar' column on every pair — the multiplication the verification harness re-does, pair by pair, from the same file this table is generated from. The rows marked 'no' are not axes the study leaves unvaried; each is counted somewhere else, and the column says where. The encoding is determined by the learner — each learner names its own encoder — so the learner row already counts the pipeline and multiplying by the encoding would count it twice; Table S7 separates the two where they can be separated. The forecast origin indexes a *second* surface on the case-study log rather than a further factor of the first, and Section 7.2 reports it. The ladder-rung row is the admissible row plus the intercept-only baseline, so the two rows are one axis at two admissibility conventions and only the admissible one enters an admissible count. And the operating point multiplies the decision-curve surface, not the scalar one, which is why the two surfaces are counted separately in Section 6.

and at how many levels, and Section 10 carries the gap. The *measure* column names which of the three declared product measures a quantity is taken under — EQUAL, REF. and CONC., placing equal, half and four fifths of the mass on the reference level — and Section 4.5 reports every decomposition under all three; the measure is a modelling choice, declared rather than left implicit (Supplement S1.2, which also says why the forecast origin and the register's quality are axes of the estimand rather than preliminaries).

3.2. What the surface estimates

Equation (1) is written on a fixed D_{train} and D_{test}, so as it stands $V_s(f)$ is a deterministic functional of the log in hand and an interval for it would cover with probability zero or one. That is not the quantity this paper reports intervals for, and the distinction has to be made before Section 4 can say what its bootstrap targets.

Definition 2 (The estimand)**.** *Let the cases of a log be a sample from a population P, and let the specification s fix the fitting procedure A_s, the cohort and eligibility rule, the split rule Π, the forecast origin, the register-quality mechanism, the baseline and the instrument. Write $A_s(P)$ for the limit of the fitted scoring rule as the sample grows under P with s held fixed. That limit needs an assumption, and it is stated here rather than where it is used: the case sequence is stationary and mixing, so that a temporal split does not separate two different populations and the moving-block bootstrap of Section 4.1 resamples a dependence structure that persists. An event log is not guaranteed to have that property. Stationarity is tested here and does not hold everywhere — the increment differs across the test half on 6 of 19 pairs (Supplement S13) — while mixing is not tested, and Section 10 carries both. The* estimand at s *is*

$$\vartheta_s(f) \;=\; m_\theta\Big(A_s(P, B \cup \{f_q\}),\, P\Big) \;-\; m_\theta\Big(A_s(P, B),\, P\Big),$$

and $V_s(f)$ of (1) is its plug-in estimate on the log.

Three things follow and are used throughout. The split rule and the pipeline are *part of the estimand*, not nuisances to be averaged away: ϑ_s is what this analyst's procedure converges to, which is why the nested bootstrap of Section 4.1 resamples the training half rather than conditioning on it, and why the split is an axis. Under misspecification ϑ_s is not what the register is worth in the world — the simulation of Section 9 separates the two and measures the gap, calling them V_{limit} and V_{oracle}. And every cell of every figure and table in this paper is an *estimate* of ϑ_s; "resolved" in Definition 3 means that the band excludes zero for ϑ_s, not that the observed increment is nonzero.

3.3. The surface, and why it is not a scalar

Given $\mathcal{S}$ and data D, the surface is the map $s \mapsto V_s(f)$ of (1), and a conventional report is one cell of it. The paper's claim is that a single cell is not a summary of the surface in any useful sense. The point estimates alone establish that: on 18 of the 19 pairs, between a tenth and nine tenths of the admissible cells are positive, so the surface's sign is not settled by the data and the design together. Whether that unsettledness amounts to a *misstatement* is a different and much harder question, and Section 6.4 answers it with three restrictions and a withdrawal rather than with a rate. There is one state in which a scalar is defensible: when the whole admissible surface lies on the same side of zero with simultaneous confidence, a positive

number is a lower bound on a robust conclusion. Section 4.6 makes that a definition.

3.4. The relative reduction, and its pathologies

Comparisons between two baselines are often stated as a proportion. Write $B_0 \subset B_1$ and

$$D_s(f) \;=\; V_s(f \mid B_0) - V_s(f \mid B_1), \quad R_s(f) \;=\; 1 - \frac{V_s(f \mid B_1)}{V_s(f \mid B_0)} \;=\; \frac{D_s(f)}{V_s(f \mid B_0)}. \tag{2}$$

D is an absolute difference in the metric's own units. R is a ratio and inherits every pathology of one: the denominator can be small, zero or negative; R is not comparable across metrics, because Remark 1 says the two are not functions of each other; and R can exceed 1 or fall below 0, so reading it as "the share absorbed" is a category error whenever it leaves $[0, 1]$.

We therefore make D and the two increments primary throughout and treat R as a secondary descriptive statistic under three rules declared before any R in this paper was computed (Supplement S3): it is reported only where $V_s(f \mid B_0)$ is resolvably positive under the *simultaneous* band; its interval is a Fieller set whose *kind* — bounded, unbounded, or the complement of an interval — is printed beside it, and of the 10,584 reductions this paper computes 5,282 have a non-bounded set where a percentile interval would report a bounded one for every single one of them; and its magnitude is never compared with its spread across metrics as though both lived on one scale.

3.5. Two facts about the metric, and why they are remarks

Two properties of the reduction are used later; Supplement S2 carries both in full, with their constructions and the richness assumption the second needs.

Remark 1 (The metric is an argument of the estimand)**.** *No function carries the AUC reduction to the average-precision reduction: two configurations can share an AUC reduction to* 2.2×10^{-16} *while their average-precision reductions differ by 0.417. The reduction is therefore not identified by the data, the feature and the two baselines, which is why every table in this paper names its instrument and why the minimal practically important difference of Section 6.4 is declared in one instrument's units and applied there only.*

The second — Remark S1, that a reduction is invariant to a declared family of recalibrations exactly when the metric is *affine* under it — is stated and proved in Supplement S2, with the 18 certificates it licenses.

4. Estimation and inference

4.1. The estimator, and uncertainty that includes the model

The pipeline is frozen and identical on every log, and the *pipeline* axis spans encodings as well as model families; every learner, encoder and calibrator is the `scikit-learn` implementation (Pedregosa et al., 2011) at the version `requirements.lock` pins. Attributes are assigned to roles by the registered rules of Section 5, cases are ordered by their first timestamp, and the split is temporal (Supplement S3.2). The usual interval for such an increment resamples test rows with the models held fixed, which excludes training-sample, model-selection, encoder and split variability and the temporal dependence between cases. Every interval here is instead a *nested moving-block bootstrap*: one draw resamples the training half by moving blocks (Künsch, 1989) of length $\lceil n^{1/3} \rceil$, refits every arm of every rung under every learner and quality mechanism, and evaluates on a moving-block resample of the test half, with encoders, frequency tables, target encodings, register rankings and calibrators rebuilt inside the draw. On the simulation of Section 9 the fixed-model interval covers a median 92.1% and the nested basic interval 93.0% — about 1.3 Monte Carlo standard errors apart, which that design cannot resolve — so the nested construction is retained because the variability it admits is variability the estimand contains, not because it measures better here; the price it charges is the subject of the next two paragraphs.
Throughout, K denotes the **cardinality of the register field** — the number of distinct values f takes on a pair, 337 at the median pair and 5,245 at the largest (Table 3) — and n the number of training cases. The ratio K/n governs everything in this subsection and in Section 6.3.

Refitting inside the draw breaks the percentile interval. A bootstrap resample holds about $1 - e^{-1}$ of a high-cardinality categorical's distinct levels, so every refit sees a sparser register and returns a smaller increment; the bootstrap distribution is displaced and a percentile interval taken from it is a correct interval for the wrong quantity. On the sparse world of Section 9 it covers 37.2% against a nominal 95%, *worse* than the fixed-model interval refitting

was meant to improve on. Every interval here is therefore the *basic* (pivotal) interval $[\,2\hat{V} - q_{1-\alpha/2},\ 2\hat{V} - q_{\alpha/2}\,]$ (Davison and Hinkley, 1997), and simultaneous bands are recentred by the same shift (Supplement S3.7).

Weights rather than subsets, and what the comparison found. Every band this paper reports comes from a bootstrap that draws Dirichlet *weights* per moving block rather than a subset of rows, so every row has positive weight in every draw and every register level is present in every refit. The multinomial alternative was run beside it over the same design, at the same draw count and with the same seeds, on 24,624 cells, and Supplement S3.8 reports the comparison in full; two findings carry into the article. Under a multinomial resample a draw can drop an arm entirely, so the cell gets no band at all and is counted as *unresolved* beside cells that were banded and straddled zero — 270 of 3,420 cells on this design, against 0 under weights. And level loss is not what displaces the bootstrap distribution: weights raise the share of the register's levels a draw contains from 80.8% to 100%, and the displacement's median moves only from 0.0050 to 0.0041, while the share of cells whose pivotal interval excludes its own estimate falls from 3.9% to 1.3% and not to zero. The displacement that remains is unexplained: the level-loss mechanism of Supplement S3.7 is the reason the percentile interval is not used, and it is not a diagnosis of the residual, which Section 10 carries as an open defect.

4.2. Three surfaces, and which one a claim may quantify over

A statement of the form *this surface is uniformly beneficial* is a statement about a denominator, and the denominator has to be the set the statement ranges over. Three sets are distinguished, named, and counted from the files rather than quoted (Table S14). The **declared surface** is every scientifically admissible specification: the product of the axis levels declared in Section 5. The **computational surface** is every specification actually evaluated; here it equals the declared surface — 29,040 scalar cells against 29,040 declared, with 0 duplicates and 0 missing — and the audit script asserts that identity rather than the manuscript asserting it. The **inference surface** is every specification carrying bootstrap draws; it is smaller for one reason, that a draw refits the whole pipeline and its cost is the declared surface's multiplied by the draw count.

The rule that selects it reads neither a result nor the log: one design on every pair — two learners, two splits, three register-quality con-

ditions and three admissible rungs, crossed, 36 cells a pair and 3,420 scalar family members across the corpus, at 400 draws — and the audit checks rather than declares that it is a complete factorial, each combination present exactly once on every one of the 19 pairs, so a corpus median over these families is a median over one design (Supplement S3.2).

A design, not a corner. Because the inference surface is a balanced factorial, the decomposition of Section 4.5 can be computed on it as well as on the declared surface — one decomposition per pair per instrument, over 36 cells each — and Supplement S14.2 reports that the axis carrying the largest first-order index agrees between the two surfaces on a minority of pairs. That is a comparison between two designs that declare different levels, and so a statement about the levels an analyst declared rather than a verdict on either surface.

What the design does not change is the thing that matters most for reading a region label. The inference family is a fraction of the admissible surface — a median 16.7% of a pair's admissible cells, and less than a twentieth on the pair with the richest grid — and **a label computed over fewer cells is systematically cleaner than one computed over all of them**. Section 4.6 says which label that most flatters, and Section 6.3 reports where it lands.

Every quantity that does *not* need a bootstrap — the decomposition, the sign-disagreement rate, the regret comparison — is computed on the full declared surface; only the region labels are computed on the inference surface.

4.3. Simultaneous bands over the family the claim ranges over

A decision curve read across 31 thresholds, a ladder read across rungs, or a surface read across every admissible cell, is a family of statements, and a pointwise interval does not control the probability that *some* member of the family is wrong. For a declared family $\mathcal{F}$ of cells with bootstrap draws $V_c^{(b)}$ and per-cell standard errors $\hat{\sigma}_c$, let

$$T^{(b)} \;=\; \max_{c \in \mathcal{F}} \frac{\left| V_c^{(b)} - \bar{V}_c \right|}{\hat{\sigma}_c}, \qquad q_{1-\alpha} = \text{the } (1-\alpha) \text{ quantile of } \{T^{(b)}\},$$

and report $\hat{V}_c \pm q_{1-\alpha}\hat{\sigma}_c$. This is the max-$t$ (Romano–Wolf) construction (Romano and Wolf, 2005) and has family-wise coverage over $\mathcal{F}$ asymptotically, and over nothing larger.

The family must be the family. Within-cell bands, each covering five members, do not cover the several hundred cells that a surface-level label ranges over. Four interval objects are declared and every table says which it prints; WHOLE-SURFACE — max-t over *every* scalar cell of the pair's inference family, a median 16.7% of its admissible scalar cells (Section 4.2) — is the only family a surface-level claim may use, and it has a median 180 members against the within-instrument family's five, at a median critical value of 5.03 against that family's 2.37 and the pointwise 1.96. Family control is per log–target pair; every cross-pair statement here is descriptive. Supplement S3.3 gives these four *interval* objects — which are not the reporting objects of Table 1 — and the rule for dropping a replicate that leaves a cell without outcome variation.

Estimating the critical value. Two estimators of $q_{1-\alpha}$ are computed on every family. The first is the empirical $(1-\alpha)$ quantile of the B observed maxima $T^{(b)}$ — the Romano–Wolf construction as cited — whose precision is that of an order statistic of B values: its order-statistic interval has a median width of 1.35 on this corpus's families. The second standardises the draws and multiplies them by independent standard normals, which gives a statistic with their empirical covariance that can be generated 20,000 times for the cost of matrix arithmetic (Chernozhukov et al., 2013), so the budget fixes B and not the precision of q (a Monte Carlo interval of width 0.036). The multiplier is an approximation: it reproduces the covariance of the draws and therefore only the excursions a Gaussian process makes, and on every family here it lies below the empirical quantile's whole order-statistic interval (Supplement S3.5). **The operative critical value is the empirical quantile**, at a median 5.03 on the whole-surface families against the multiplier's 3.36; Section 9.2 is the measurement that decides it, and every table prints what the multiplier would have resolved beside what the reported band does. A max-t band also inherits the coverage of the interval it is built from, which Section 9.1 shows is governed by the ratio of register cardinality to training size; Supplement S9 derives a widening calibrated at each pair's own ratio and reports it as a sensitivity.

What these bands cover, measured against a known answer. Section 9.2 measures family-wise coverage over synthetic families matched to this corpus in size, draw count, cross-cell correlation and tails, whose truth is known by construction and whose bootstrap is *ideal*, so every coverage it reports is an

upper bound on what the real procedure attains. On the family matched to the design this paper reports — 180 cells at 400 draws under this corpus's tails — the empirical quantile's band covers 94.7% against a nominal 95%, at a Monte Carlo standard error of 0.5%, and 94.6% under a non-zero truth, the regime in which a cell can be resolved correctly and a region label is actually being asked to hold. The multiplier's band covers 91.5% and 92.6% on the same families, several standard errors short, and no draw budget closes that gap: along a ladder of draw counts its coverage plateaus from 150 draws on, while the empirical quantile's rises to its level by 400. **The level is bought with width**: the empirical band is 1.24 times as wide on the matched family and 1.50 times on the corpus's own families, and it resolves 698 cells corpus-wide where the multiplier would resolve 890 (Section 6.3). What the measurement does not cover is stated with it: the bootstrap is ideal there, so the K/n-governed shortfall of the pointwise interval underneath (Section 9.1) and whatever the real resampling adds are outside it, and Section 10 carries both.

4.4. Planned contrasts, and a p-value the design can produce

Five contrasts on the primary log were fixed before this analysis round — that the register adds to the intake baseline, that it still adds once the free field is admitted, that it adds at the later forecast origin, that the absorption D is positive, and that a half-empty register is worth less than a full one — each with its complete estimand in Table S15. They are not confirmatory, because they were fixed after the data were seen, so their Holm-adjusted one-sided bootstrap p-values (Holm, 1979) are reported for legibility: 5 reject at $\alpha = 0.05$, but a contrast is called resolved only when its interval says so, and on PC3 the one-sided p-value and the two-sided basic interval disagree, so the count under this paper's own rule is 4 (Supplement S3.4, which also gives the smallest attainable adjusted value, 0.00250 at 2,000 draws). Every other comparison in this paper is *exploratory* and carries no multiplicity correction, because none would be meaningful over a surface enumerated in full.

4.5. The sensitivity decomposition

Because the surface is a complete factorial over its axes — the audit of Section 4.2 establishes that it is, cell for cell — the variance of V_s admits an exact and *orthogonal* functional-ANOVA decomposition (Sobol', 2001; Saltelli et al., 2010) under a product measure $\mathbb{P} = \bigotimes_i w_i$, with first-order and total indices S_i and S_{T_i} computed exactly rather than estimated (Supplement S3).

The closest precedent is Hutter et al. (2014), who decompose an algorithm's performance over its *configuration* space by the same device; here the axes are analysis decisions, the components are exact over a complete factorial rather than estimated from a surrogate, and the decomposition is one of three reporting objects rather than the conclusion.

The split is an error stratum, and the primary decomposition treats it as one. Five of the split axis's six levels are expanding-origin folds on the same data. They are different *samples*, not different analyses, and a decomposition that pools them reports sampling variability as specification sensitivity. Because the decomposition is exact and orthogonal, the repair is a partition rather than a model: every component either involves the split or does not, so

$$1 \;=\; \underbrace{\textstyle\sum_{\emptyset \neq u \subseteq \mathcal{A}} S_u}_{\text{analyst choice}} \;+\; \underbrace{S_{\{\text{quality}\}}}_{\text{counterfactual}} \;+\; \underbrace{\textstyle\sum_{u \not\ni \text{split}} S_u - (\cdots)}_{\text{mixed}} \;+\; \underbrace{\textstyle\sum_{u \ni \text{split}} S_u}_{\text{resampling}}, \qquad (3)$$

where $\mathcal{A} = \{\text{pipeline}, \text{rung}\}$ and the third term is what the split-free components carry beyond the first two, and the four terms are computed, not estimated (Table 5); the identity holds to 4.4×10^{-16}. The first three are what survives fold-averaging, and they are decomposed again *on* the fold-averaged surface, where each index is a share of the variance that fold-averaging leaves and the higher-order share is net of resampling. **Section 6.2 reports that decomposition as the primary one** and the pooled decomposition beside it, because the pooled one is what a reader of the earlier specification-curve literature will expect and the difference between them is itself a result.

The per-axis difference $S_{T_i} - S_i$ is the variance axis i is *involved* in beyond its main effect. These overlap — a two-way component appears in both its axes' total effects — so their largest member is not the share carried by interactions. That share is

$$S_{\text{int}} \;=\; 1 - \sum_i S_i, \qquad (4)$$

which this paper computes together with every disjoint component, so the identity $\sum_u S_u = 1$ is asserted numerically rather than assumed (Supplement S3). The other quantity is retained as the *largest per-axis interaction involvement*; the two differ substantially, at 43.7% against 56.0% at the median pair.

4.6. Resolution regions and the robustness index

Definition 3 (Cell labels and surface regions)**.** *Let $\mathcal{F}$ be the declared inference family of Section 4.2 for a pair — the cells that carry bootstrap draws, a median 16.7% of that pair's admissible scalar cells. Under the simultaneous band of Section 4.3, a cell of $\mathcal{F}$ is* beneficial *if its lower band is above zero,* harmful *if its upper band is below zero, and* unresolved *otherwise. A surface is* uniformly beneficial *if every cell of $\mathcal{F}$ is beneficial;* conditionally beneficial *if some are and none is harmful;* uniformly harmful *and* conditionally harmful *in mirror image;* sign-changing *if there is at least one cell of each kind; and* unresolved *if none is resolved — **or if fewer than 5% of $\mathcal{F}$ is resolved at all**. The* robustness index *is*

$$\rho \;=\; \frac{\#\{\textit{beneficial}\} - \#\{\textit{harmful}\}}{|\mathcal{F}|} \;\in\; [-1, 1],$$

and every directional label is reported with the counts of beneficial, harmful and unresolved cells it rests on.

The minimum resolved share is not free, and the family the label is relative to is not neutral. Without the share a surface earns a direction from as few as two resolved cells of a hundred and eighty, which on a family the data barely touch is a statement about the band's width and not about the register; 3 of the 9 pairs that carry a direction lose it under the condition (Table 6), and ρ is printed beside the triple everywhere because ρ alone cannot distinguish two resolved cells from ninety. The threshold is a declared convention: at 1% it withdraws 0 directions, at the declared 5% 3, at 10% 5, and Table 6's source file carries all three. And a label computed on a sixth of a pair's cells is systematically cleaner than one computed on all of them, with *uniformly beneficial* the label most helped, so the manuscript never says a surface is uniformly beneficial without this restriction being in force, and Section 6.3 reports the one pair that reaches it together with the share of its admissible cells the label was earned over.

The two harmful labels are not decoration, and what this corpus does with them is a case of the minimum share doing its work. Table 6, which prints the label before and after the minimum share, carries 2 conditionally harmful and 0 uniformly harmful surfaces out of 19 before it; each conditionally harmful label rests on at most 4 resolved cells of 180, so Definition 3 withdraws them, and under the definition this paper reports the count of either kind is 0. The label is available and the data do not earn it. $\rho = 1$ is the only state in

which one positive number is a safe summary of a surface, and the empirical question this paper answers is how often it obtains. The region is defined over the *scalar* family; folding the decision-curve cells into it would let a single extreme threshold decide the label of a whole surface, so the decision curve gets its own region (Section 8). The intercept-only rung is excluded from every admissible set, because a reduction measured against a model with no features is inflated by construction; the master surface file *does* carry that rung, which is why the two counts in Table S14 differ.

4.7. Specification regret, as a robustness check

This subsection defines a check and not a reporting object. What *specification regret* reports is a null — the one-number rule's median excess regret is zero in every instrument, the out-of-sample ordering reverses under rolling folds (Section 6.6), and on calibrated net benefit what separates the rules is the conservative one rather than the one-number one (Section 8.3) — and a null is a check on the paper's other two arguments rather than a result of its own. It is reported in full, and it is not a contribution.

The construction asks what choosing one specification costs against the best one, on a scale a decision can use; the identity map's maximum excess is exactly zero. Supplement S3.1 gives the three declared designs — metric-specific, calibrated net benefit, and partially identified over a declared set of monotone utility maps — and the four decision rules compared under them.

4.8. A reporting standard, in one paragraph

Report the design space and the measure over it; the denominator the claim quantifies over, named as one of the three; the absolute increment at every rung rather than a ratio; the decomposition with $1 - \sum_i S_i$ reported as such; the region and ρ under the band's own family, with the counts they rest on; and the reduction R last if at all, and only where its denominator is resolvably positive. Supplement S21 states the standard in full, together with the errors this study made before it got them right, a six-step recipe, the `fieldvalue` package (Anonymous, 2026) that implements the objects and refuses by construction to emit a single number, and a worked example on a dataset that is not an event log at all.

5. The registered benchmark design

5.1. What was fixed, and when

The rules that turn an arbitrary event log into an instance of this problem were committed before any result on a new log existed, in `PROTOCOL.md` for the corpus and `AUDIT-PROTOCOL-2.md` for the practice pilot; the version history shows the registering commit adding one file and no result file, and six amendments were declared afterwards, each with its date, its cause and the result it changed (Supplement S1). **This is not a timestamped third-party registration**: what is offered is a file in a repository whose history the author controls, so any claim that a rule was fixed in advance is a claim about the repository and no stronger than the repository. What the design buys is that the rules are legible, dated and amendable only on the record, not that an outside party watched them being written.

5.2. Roles

A log is admitted if the registered rules can find three things in it without reading an outcome.

The register f. A high-cardinality attribute that is not resource-like, has at least 50 distinct values, is reused across at least two cases on average, and — by amendment 1 — whose values recur across the temporal split, so that a document number is not mistaken for a maintained entity.

The free field g. A resource-like attribute, present on at least 50% of cases, whose value varies within a case, so that it can carry information about routing. Only the *first* value it takes on a case is used as a feature, while the handover target is defined from the whole sequence of values — so the feature is what is known when the case opens and the target is what happens afterwards, and the field cannot inform the model about its own future. The distinction is not cosmetic: the same attribute supplies both, and using the case-level set would put the answer in the features.

The intake block B_0. Every remaining attribute of cardinality at most 200 that is neither an outcome, nor a timestamp, nor a relabelling of the activity, nor a deterministic coarsening of f.

log	target	ITSM	register field f	free field g	axes: pipe/split/qual/adm. rungs	cases	levels of f
BPIC13_closed	handover	yes	product	org:resource	2/6/6/3	1,487	337
BPIC13_incidents	duration	yes	product	org:resource	2/6/6/3	7,554	704
BPIC13_incidents	handover	yes	product	org:resource	2/6/6/3	7,554	704
BPIC14	duration	yes	CI Name (aff)	assignment group	4/6/10/4	46,606	3,019
BPIC14	handover	yes	CI Name (aff)	assignment group	4/6/10/4	46,606	3,019
BPIC15_1	duration	–	case:parts	org:resource	2/6/6/3	1,199	89
BPIC15_1	handover	–	case:parts	org:resource	2/6/6/3	1,199	89
BPIC15_3	duration	–	case:parts	org:resource	2/6/6/3	1,409	94
BPIC15_3	handover	–	case:parts	org:resource	2/6/6/3	1,409	94
BPIC15_4	duration	–	case:parts	org:resource	2/6/6/3	1,053	53
BPIC15_4	handover	–	case:parts	org:resource	2/6/6/3	1,053	53
BPIC15_5	duration	–	case:parts	org:resource	2/6/6/3	1,156	107
BPIC17	duration	–	case: RequestedAmount	org:resource	2/6/6/3	31,509	701
BPIC19	duration	–	case:Vendor	org:resource	3/6/6/3	251,734	1,975
Helpdesk	duration	yes	customer	Resource	2/6/6/3	4,580	394
RoadFines	duration	–	article	org:resource	3/6/6/3	150,370	66
Sepsis	duration	–	Diagnose	org:group	2/6/6/3	1,050	147
UCI498	duration	yes	caller_id	assigned_to	2/6/6/3	24,918	5,245
UCI498	handover	yes	caller_id	assigned_to	2/6/6/3	24,918	5,245

Table 3: **Roles and levels, per pair.** Which recorded attribute plays the register f and which the free field g on each admitted pair, whether the pair is IT service management, and how many levels each varied axis carries: pipeline, split, register-quality condition, admissible baseline rung — the intercept-only rung is in the surface and in no admissible set, and is not counted here. Every pair also carries the 5 scalar instruments, so the per-pair factorial is 5 axes wide and not the nine of Definition 1; Section 5 says which axes are fixed on which pairs and why. 'cases' is the admitted case count and 'levels of f' the register's cardinality. The master table cannot be read without it.

Coarsenings of f — its type, its subtype, the service it belongs to — are given their own role rather than being swept into B_0, because which *layer* of a register pays is a separate question, answered in Section 7.4.

Table 3 is what the rules found, and it is the design space as it was actually varied: per pair, the attribute playing each role and the number of levels each axis carries. Two things are visible there that Definition 1 does not say. The *target* is not an axis of any surface — each of a log's two targets has a surface of its own, and the cohort × target design exists only in Section 7.1 — and the per-pair factorial is 5 axes wide, not the nine of the definition: the forecast origin varies only on the case study's log, the operating point only in the decision-curve family, and the cohort only in Section 7.1. Section 10 carries the difference as a limitation.

5.3. Targets, splits and exclusions

Two targets are defined for every log — *handover*, whether more than one resource group touches the case, and *duration*, whether it runs longer than the training half's median. Cases are ordered by first timestamp and split 70/30, with five expanding-window rolling-origin folds beside it, and fields determined only at closure are excluded by name (Kaufman et al., 2012; Kapoor and Narayanan, 2023; Weytjens and De Weerdt, 2022). A pair is excluded only for a registered reason, and every admitted pair is analysed and reported — running the surface only where the reduction resolves would condition the analysis on the outcome. Supplement S1.3 gives the rules, the exclusion codes and the admission margin at the upper edge of the prevalence window.

Two exclusions a reader of Table 3 will ask about. BPIC15_2 is excluded under SMALL_N at 832 cases while its four siblings are admitted — the rule is a floor on the case count and nothing about the log's content — and the further BPIC15 targets, the BPIC20 logs and BPIC12 are excluded under NO_HEADROOM or NO_G, which are prevalence and role failures. And the registered handover target on the case study's log sits at the upper edge of the admission window: test prevalence 0.927 against a window that closes at 0.95, which excludes other logs at 0.959. That pair carries 4,800 of the corpus's 29,040 admissible cells — the log carries 9,600 across its two targets — so a corpus statistic pooled over cells is partly a statement about it; Supplement S14.17 reports every headline on the IT service management (ITSM) family separately for this reason, and Section 10 asks whether the window should be tightened.

5.4. What the corpus does and does not have in common

The corpus is 13 public logs spanning 6 domains, and it is heterogeneous: this is a *benchmark of specification sensitivity across heterogeneous prediction problems*, not a replication of one finding on several organisations, so no effect magnitude is averaged across domains as though it estimated one universal register value, and the 8 ITSM-like pairs are analysed as a group as well. Supplement S16 names, for every pair, the domain, the field playing each role, the target's interpretation and the construct threat that pair carries.

6. Multi-log results

Every point estimate in this section comes from one master surface — one file, named in the code-availability statement — computed by one pass of one pipeline over 19 log–target pairs, 4 learners, 6 splits, 10 register-quality conditions, the nested baseline ladder and 36 instruments including the 31-point decision-curve grid. Its 273,024 rows *include* the intercept-only rung, so that the surface is complete; no admissible set contains it, and the 29,040 admissible scalar cells every headline is computed on exclude it. Every interval is the basic construction of Section 4.1 and every band the whole-surface simultaneous construction of Section 4.3, and each table names which it prints. The denominator audit is in Table S14: declared equals computed on all 19 pairs, with 0 duplicates and 0 missing, and the inference surface (3,420 scalar members) is a subset of it cell for cell, which `verify_numbers` asserts.

6.1. The master table

Table 4 is the frozen master table: one row per log–target pair, the increment at the reference specification with its simultaneous band, the region label, the robustness index and the sign-disagreement rate of a conventional one-number report. The reference specification is the same everywhere — one-hot logistic regression, the single temporal holdout, the clean register, the intake block plus the free field as the baseline, ROC AUC — and its region and ρ columns are the ones Section 6.3 reports. The reference cell's sign and the region label need not agree, and on several pairs they do not: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative, because the label ranges over the whole inference family and the reference cell is one member of it. Table 6 prints the counts each label rests on, and Table 3 which attribute plays the register on each row.

6.2. What the choices are worth

Table S8 and Figure 3 give the decomposition, computed within each instrument in that instrument's own units, under the equal-level measure. The table's first four columns are, per pair, the median over the five instruments of each axis's first-order index; its last column is the largest per-axis interaction *involvement*. Three aggregations of this table are used here and in Supplement S14.11 and they are not interchangeable, so each is named where it appears: the median over pairs of a column of this table; the 28.7%

log	target	V at reference	pointwise	simultaneous	region	resolved share	ρ	sign-disagreement rate
BPIC13_closed	handover	−0.009	[−0.042, +0.018]	[−0.074, +0.052]	unresolved	0.0	0.000	0.284
BPIC13_incidents	duration	0.028	[+0.009, +0.055]	[−0.016, +0.081]	conditionally beneficial	6.7	0.067	0.393
BPIC13_incidents	handover	0.037	[+0.028, +0.049]	[+0.015, +0.062]	conditionally beneficial	40.0	0.400	0.111
BPIC14	duration	0.096	[+0.089, +0.108]	[+0.079, +0.119]	uniformly beneficial	100.0	1.000	0.106
BPIC14	handover	0.167	[+0.146, +0.207]	[+0.093, +0.250]	conditionally beneficial	95.0	0.950	0.244
BPIC15_1	duration	0.001	[−0.036, +0.036]	[−0.103, +0.098]	unresolved	0.0	0.000	0.423
BPIC15_1	handover	0.007	[−0.007, +0.020]	[−0.033, +0.049]	unresolved	0.0	0.000	0.242
BPIC15_3	duration	0.035	[−0.006, +0.068]	[−0.055, +0.126]	unresolved (below minimum share)	1.1	0.011	0.191
BPIC15_3	handover	−0.003	[−0.050, +0.052]	[−0.126, +0.122]	unresolved (below minimum share)	2.2	−0.022	0.616
BPIC15_4	duration	0.035	[+0.009, +0.079]	[−0.055, +0.147]	unresolved	0.0	0.000	0.623
BPIC15_4	handover	0.001	[−0.021, +0.042]	[−0.093, +0.106]	unresolved	0.0	0.000	0.618
BPIC15_5	duration	0.032	[−0.020, +0.103]	[−0.146, +0.211]	unresolved	0.0	0.000	0.328
BPIC17	duration	0.002	[−0.004, +0.013]	[−0.018, +0.026]	unresolved	0.0	0.000	0.353
BPIC19	duration	0.024	[+0.011, +0.040]	[−0.020, +0.071]	sign-changing	45.0	−0.339	0.360
Helpdesk	duration	−0.020	[−0.045, −0.002]	[−0.069, +0.021]	unresolved (below minimum share)	1.1	−0.011	0.257
RoadFines	duration	0.031	[+0.024, +0.038]	[+0.013, +0.050]	sign-changing	32.2	0.256	0.173
Sepsis	duration	0.153	[+0.102, +0.199]	[+0.059, +0.248]	conditionally beneficial	56.7	0.567	0.056
UCI498	duration	−0.007	[−0.014, +0.005]	[−0.023, +0.015]	conditionally beneficial	7.8	0.078	0.493
UCI498	handover	−0.003	[−0.007, −0.001]	[−0.012, +0.004]	unresolved	0.0	0.000	0.481

Table 4: The master table. One row per admitted log–target pair. 'pointwise' is the 95% interval at the reference cell, basic (pivotal) construction; 'simultaneous' is that same cell's *whole-surface* simultaneous band at the empirical critical value of Section 4.3, the construction Section 9.2 measures at its nominal level on the family matched to this design. They are different objects, and the region label uses only the second. Then the resolution region and the robustness index ρ. **No** (n, K) **calibration is applied**: that factor widens the pointwise interval underneath the band and is reported as a sensitivity in Table S9, which also prints what the multiplier approximation would have resolved. **The region is the label Definition 3 yields**, minimum resolved share included: 'resolved share' is the percentage of the inference family the band resolves, and a direction is withheld below 5% of it, which withdraws the direction on 3 of the 9 pairs that carry one. Table 6 marks 3 rows: a pair the band already left unresolved is not marked, and 0 of the marked rows are sign-changing, which carry no direction to withdraw. Last, the share of admissible specifications whose sign disagrees with a conventional one-number report, under the equal-level measure, which is the 'all-cells rate' column of Table S10. The reference cell's sign and the region need not agree, and on several pairs they do not: a surface can be conditionally beneficial while the one cell an analyst would have stood on is negative.

figure, which is the median over pairs of the median over instruments of the *per-instrument* largest first-order index, and which is therefore not obtainable by taking a maximum across the four printed columns; and the corpus medians of the higher-order share. The per-instrument quantities the second is built from are in Table S26.

Most of what the pooled surface varies over is not specification at all. Table 5 partitions the decomposition by (3), exactly and on every decomposition separately. Taking each column's median across the corpus — *four medians, which do not add, and are not four parts of one pair* — the components involving the split carry 71.7% of the variance, against 13.3% for the choices an analyst makes, 4.2% for the register-quality counterfactual and 4.5% for the components that join them. Resampling is the largest of

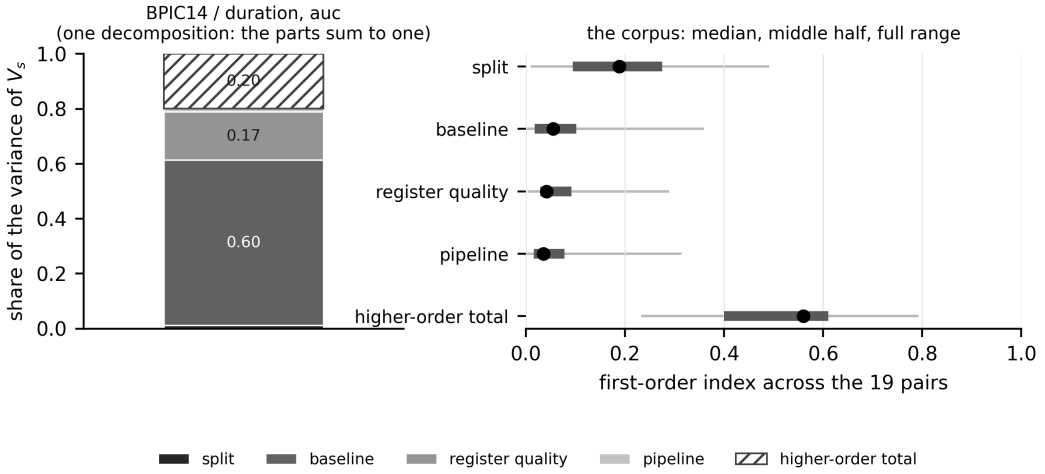


Figure 3: The decomposition, within instrument, under the equal-level measure. *Left*: one pair at one instrument, so the components are a decomposition and the parts sum to one exactly — the largest pair in the corpus, at the reference instrument, which is the cell every other table reports at. The hatched segment is the total higher-order share $1-\sum_i S_i$ of (4). *Right*: each axis's first-order index across the corpus, as the median over pairs with the range the middle half occupies and the full range. Nothing on the right is stacked, because a median over pairs and a median over instruments do not add and a stacked bar of them would invite the addition.

the four on 17 of the 19 pairs. Dropping the one split level that is a genuine analyst choice — the single temporal holdout, leaving five folds of one rule — moves that share only from 71.7% to 71.0%, so the stratum is not an artefact of mixing a rule with its folds. The two exceptions are the case study's own pairs, where the analyst axes carry the richest grids in the corpus — 4 pipelines and 10 register-quality conditions, against at most 3 and 6 on every other pair (Table 3) — and the ordering reverses.

Net of resampling, no single axis dominates — and this is a weaker statement than the pooled surface makes it look. On the fold-averaged surface, which carries 28.3% of the pooled variance, the median first-order index is 21.9% for the baseline rung, 19.2% for the register-quality condition and 15.2% for the pipeline, and at the median pair the largest first-order index *within an instrument* is 38.9% — a median over pairs of a median over instruments of the per-instrument largest, and so not the largest of the three medians just quoted. The total higher-order share is 26.3% once resampling is removed, against 56.0% on the pooled surface, and it no longer

log	target	analyst choice	counterfactual	mixed	resampling
BPIC13_closed	handover	31.5	2.9	5.8	59.8
BPIC13_incidents	duration	13.3	2.9	3.0	79.6
BPIC13_incidents	handover	19.2	9.9	4.5	42.3
BPIC14	duration	71.9	15.1	7.9	5.1
BPIC14	handover	73.9	12.7	10.3	3.6
BPIC15_1	duration	9.4	3.9	4.7	80.5
BPIC15_1	handover	13.4	3.6	3.6	62.6
BPIC15_3	duration	11.7	8.1	1.5	77.6
BPIC15_3	handover	22.3	0.4	2.5	76.1
BPIC15_4	duration	11.3	2.5	5.6	81.6
BPIC15_4	handover	12.4	3.1	5.7	71.7
BPIC15_5	duration	17.4	4.2	3.9	69.9
BPIC17	duration	4.2	6.2	5.2	84.1
BPIC19	duration	3.8	1.1	2.8	90.9
Helpdesk	duration	31.7	8.6	1.7	53.6
RoadFines	duration	7.0	28.9	2.1	61.7
Sepsis	duration	13.2	14.5	5.0	71.9
UCI498	duration	11.0	3.0	4.3	81.1
UCI498	handover	41.0	7.1	9.3	42.4

Table 5: **The decomposition with the split as an error stratum.** Per cent of the pooled variance of V_s, per pair, median over the 5 instruments, under the equal-level measure. The four columns are an exact partition of the orthogonal decomposition's components, and on *each* of the 304 decompositions they sum to 100 to within 4.4×10^{-16}. **The printed rows are medians over instruments and therefore need not sum to 100**, for the same reason Figure 3's right panel is not stacked: a median is not linear. 'analyst choice' collects every component built only from the pipeline and the baseline rung, 'counterfactual' the register-quality main effect, 'mixed' the components that join the two, and 'resampling' every component involving the split — five of whose six levels are expanding-origin folds on the same data. Dropping the sixth, the single temporal holdout, so that the axis is folds and nothing else, moves the last column's corpus median from 71.7% to 71.0%.

exceeds the largest first-order index: pooled, the interactions carry more than any main effect on 13 of 19 pairs; net of resampling, on 7. *Most of the variance belongs to no single axis* is a property of a surface that counts five folds as five specifications, and it does not survive treating them as what they are. What does survive is the weaker and still sufficient claim: which axis leads is a property of the pair — the rung on 8, the register-quality condition on 5 and the pipeline on 6 — so an analyst cannot learn from somebody

else's paper which of their own choices their answer turns on.

The pooled decomposition is reported beside it because it is what the specification-curve literature computes, and the difference between the two is this subsection's result; Supplement S14.11 gives its four first-order medians, the largest per-instrument first-order index, the higher-order share with its spread, and the largest per-axis *involvement*, which is a different quantity again. The measure is not neutral: across the three product measures the higher-order share moves by 24.3%, and under the measure concentrated on the reference cell the interactions no longer carry more than the largest first-order index, so *the interactions carry more than any main effect* is a claim about a measure and not only about a surface (Supplement S14.11).

The baseline axis moves the answer by more than the answer, on the pairs where an analyst has more than one baseline to choose between. Holding every other axis at the reference and varying only the baseline rung, the increment moves by 0.073 AUC at the median pair and 0.252 at the widest — larger than the increment itself on 15 of the 19 pairs. The half-of-intake rung carries that range at the median pair and not at the widest, and it is impoverished in the way the excluded intercept-only rung is, so the spread is also reported without it: 0.0055 AUC at the median pair, 0.2495 at the widest, exceeding the reference increment on 7 pairs. On 17 of the 19 pairs only two rungs then remain, so that spread *is* the absorption D of the free field, which Section 7.5 reports with its own interval (Supplement S14.19 gives the per-pair fall). The index and the spread answer different questions — a share of the variance while every axis varies, against how far one axis moves the answer alone — and both are reported because either alone misleads.

The pipeline axis is two axes, and it does not run at full width everywhere. Its levels are a model family together with an encoding and, in one case, a calibration; on the declared surface the case study's log carries 4 levels, the two largest logs 3 and every other pair 2, by a rule that reads the log's row count and nothing else (Table 3), while inside the inference family every pair carries the same two levels, differing in model family. Crossed on the case study's log — both families against all three encodings, 6 complete factorial pipelines on each target — the encoding is the larger of the two, 10.1% of the variance against the family's 5.2%, with their interaction the largest two-way term. On the 8 ITSM pairs that finding travels but does not hold everywhere: the encoding leads on 6 and the family on 2, and not narrowly — which axis

leads is a property of the pair (Supplement S14.1). The remaining pairs are not crossed, which Section 10 records.

6.3. Resolution regions, and the coverage they have

Table S16 and Figure S4 carry the region labels and the robustness index for every pair, computed on the inference surface under the whole-surface simultaneous band at the empirical critical value of Section 4.3. That family has a median 180 members per pair and a median critical value of 5.03, against 2.37 for a within-instrument family and 1.96 pointwise; under the narrower family the same draws give a different region label on 12 of the 19 pairs, and the whole-surface family resolves 698 cells against the narrow family's 1,100.

What the choice of critical value costs, and what it buys. The multiplier approximation, which every table prints beside the reported band, would resolve 890 cells at a median robustness index of 0.039 against 0.000 here, with 2 pairs resolving nothing against 8; Section 9.2 measures it at 91.5% family-wise coverage on the family matched to this design and the reported band at 94.7%, so the 192 cells the reported band leaves unresolved are the resolution the multiplier was claiming without its level. Two shortfalls remain and are stated rather than corrected. Section 9.1 shows that the pointwise interval underneath the band is governed by K/n and undercovers where this corpus lives, the median pair sitting at 0.106; Supplement S9 derives an inflation factor $c(K/n)$ for that channel — a monotone regression through measured factors, log-linear slope 0.102 (standard error 0.005), which would supply 1.18 to 1.77 on this corpus — and reports the labels under it as a sensitivity, because it is fitted pointwise, applied simultaneously, and its transfer to real logs is an assumption. And the coverage measurement's bootstrap is ideal, so whatever the real resampling adds is outside it.

Of the 11 pairs that resolve anything, 9 resolve cells of one sign only — 7 beneficial and 2 harmful — and 2 resolve cells of both. That is the informative reading of the region column, and it is the one to carry away: on 9 of the 11 pairs where the data determine any sign at all, every sign they determine points the same way; on the remaining 2 they do not.

One surface of the 19 is uniformly beneficial, over a family that is 3.75% of that pair's admissible cells. Section 4.6 poses the question this answers: $\rho = 1$ is the only state in which one positive number is a safe summary of a surface, and it obtains 1 time here, on the pair carrying the

log	target	family size	beneficial/harmful/unresolved	resolved share	region	label under minimum share
BPIC13_closed	handover	180	0/0/180	0.0	unresolved	–
BPIC13_incidents	duration	180	12/0/168	6.7	conditionally beneficial	–
BPIC13_incidents	handover	180	72/0/108	40.0	conditionally beneficial	–
BPIC14	duration	180	180/0/0	100.0	uniformly beneficial	–
BPIC14	handover	180	171/0/9	95.0	conditionally beneficial	–
BPIC15_1	duration	180	0/0/180	0.0	unresolved	–
BPIC15_1	handover	180	0/0/180	0.0	unresolved	–
BPIC15_3	duration	180	2/0/178	1.1	conditionally beneficial	unresolved (below minimum share)
BPIC15_3	handover	180	0/4/176	2.2	conditionally harmful	unresolved (below minimum share)
BPIC15_4	duration	180	0/0/180	0.0	unresolved	–
BPIC15_4	handover	180	0/0/180	0.0	unresolved	–
BPIC15_5	duration	180	0/0/180	0.0	unresolved	–
BPIC17	duration	180	0/0/180	0.0	unresolved	–
BPIC19	duration	180	10/71/99	45.0	sign-changing	–
Helpdesk	duration	180	0/2/178	1.1	conditionally harmful	unresolved (below minimum share)
RoadFines	duration	180	52/6/122	32.2	sign-changing	–
Sepsis	duration	180	102/0/78	56.7	conditionally beneficial	–
UCI498	duration	180	14/0/166	7.8	conditionally beneficial	–
UCI498	handover	180	0/0/180	0.0	unresolved	–

Table 6: **What a region label rests on.** $|\mathcal{F}|$ is the inference family's size; the triple counts its cells the whole-surface band calls beneficial, harmful and unresolved. A directional label follows from as few as two resolved cells, and the last column applies the minimum resolved share of Definition 3 — 5% of the family — which withdraws the direction on 3 of the 9 pairs that carry one. A dash means the label is unchanged.

richest declared grid — over an inference family of 180 cells against that pair's 4,800 admissible ones, the smallest share in the corpus, which 2 pairs attain jointly. The label requires every cell of *that* set to resolve beneficial, and a claim about every cell is correspondingly easier to sustain over fewer of them, so the one case in which a single number is safe is a case where the claim ranges over 3.75% of the specifications an analyst could have chosen: a statement about a denominator rather than a counterexample to one. It is not an artefact of the critical value: the label holds under the multiplier band, under the empirical band this paper reports, under that band's order-statistic upper end, and under every widening the (n, K) calibration could ask for (Supplement S9.7) — the same pair every time. Supplement S14.18 gives the whole reckoning.

6.4. What one number costs

A conventional report — one pipeline, the clean register, the single hold-out, ROC AUC — disagrees in *sign* with a median 32.8% of the other admissible specifications under the equal-level measure. Three things are wrong with reading that as what a one-number report exposes a reader to, and each correction moves the denominator the next is computed on, so the rate that survives is the one taken under all three together. **The axes are not all the same kind of thing.** Table S20 partitions the decomposition into *analyst latitude* (the pipeline and the baseline rung), *resampling* (the split,

five of whose six levels are folds on the same data) and the *counterfactual* (the register-quality condition, a different state of the world): on the primary scale, which decomposes inside each instrument, analyst latitude carries a median 11.1% of the variance, resampling 18.9% and the counterfactual 4.2%, with resampling the larger on 10 of the 19 pairs; on the headroom scale, which makes the instrument a sixth axis, the ordering reverses (19.2%, 9.4% and 2.7%, resampling larger on 7 pairs), and neither is a corpus fact independent of the scale. Restricted to the analyst-latitude cells — the pipeline by the admissible rungs by the 5 instruments, register clean, split at the single holdout, a median 30 cells per pair — the median disagreement rate is 20.0% against the all-cells 32.8%: the first is what a one-number report exposes a reader to, the second what a claim ranging also over degraded registers and resampled splits is exposed to. Varying the reference one axis at a time over every level that axis carries — 26 variants in all, Table S21 — the corpus median runs 32.8% to 38.4% against the declared 32.8%, so the headline is not an artefact of which cell we called conventional.

And most cells are not cells whose sign the data determine. The median pair resolves 2 cells of the 180 it carries, and sign agreement between two draws from a distribution centred near zero is one half by construction. **And a sign flip between two increments that both round to zero is not a misstatement anybody would act on**: 8 of the 19 reference increments are themselves inside a hundredth of an AUC point, so a *minimal practically important difference* is declared — MPID = 0.01 AUC — and a disagreement is counted only when the reference increment and the cell's own increment both exceed it, on the AUC sub-surface, because Remark 1 forbids carrying an AUC threshold to another instrument (Table 7).

On the cells the corpus can resolve, the rate is 13.0%, and it is not an AUC statement. Of the 698 cells whose whole-surface band excludes zero, 91 disagree in sign with the reference cell. Split by instrument that rate runs 2.6% on ROC AUC to 20.3% on log-loss skill: the pooled figure is carried by the instruments that are not the one the reference cell is read in, which is Remark 1 showing up in the corpus.

Imposing every correction this section's own argument asks for leaves 31 cells, of which the number that disagrees is 0. Restrict to the axes an analyst chooses, to the cells the band resolves, and to increments that clear the MPID at both ends, all on the AUC sub-surface. The set that is being restricted is the inference family and not the admissible surface: only 684 of the 5,808 admissible AUC cells carry bootstrap draws, and a cell

log	target	V at reference	above MPID	ref. resolved	all cells	above MPID rate	resolved	resolved and above MPID	mean deviation, AUC
BPIC13_closed	handover	−0.0091	–	–	39.4	–	–	–	–
BPIC13_incidents	duration	0.0278	yes	–	21.8	16.2	0.0	0.0	0.0692
BPIC13_incidents	handover	0.0373	yes	yes	8.3	5.5	0.0	0.0	0.0510
BPIC14	duration	0.0959	yes	yes	8.8	0.9	0.0	0.0	0.0204
BPIC14	handover	0.1673	yes	yes	23.3	21.4	0.0	0.0	0.0623
BPIC15_1	duration	0.0011	–	–	38.4	–	–	–	–
BPIC15_1	handover	0.0071	–	–	22.7	–	–	–	–
BPIC15_3	duration	0.0354	yes	–	7.4	4.8	0.0	0.0	0.0876
BPIC15_3	handover	−0.0030	–	–	74.1	–	–	–	–
BPIC15_4	duration	0.0348	yes	–	56.5	56.3	–	–	–
BPIC15_4	handover	0.0015	–	–	70.8	–	–	–	–
BPIC15_5	duration	0.0320	yes	–	21.8	17.6	–	–	–
BPIC17	duration	0.0021	–	–	23.1	–	–	–	–
BPIC19	duration	0.0244	yes	–	17.6	16.4	0.0	0.0	0.0600
Helpdesk	duration	−0.0195	yes	–	37.5	32.4	–	–	–
RoadFines	duration	0.0306	yes	yes	4.0	2.9	0.0	0.0	0.0118
Sepsis	duration	0.1527	yes	yes	0.0	0.0	0.0	0.0	0.0402
UCI498	duration	−0.0069	–	–	61.1	–	100.0	–	0.0699
UCI498	handover	−0.0033	–	–	54.2	–	–	–	–

Table 7: **Sign disagreement once magnitude is required.** On the AUC sub-surface, because the minimal practically important difference is declared in AUC and Remark 1 forbids carrying it to another instrument. The four rate columns are per cent of cells whose sign disagrees with the reference cell: over every admissible AUC cell; over those where the reference increment and the cell's own increment both exceed the MPID of 0.01 AUC; over those the whole-surface band resolves; and over cells that satisfy both restrictions. A dash is not a zero — it is a restriction that leaves no cell to compute a rate on. The last column is a magnitude and not a rate: the mean distance from the reference increment over the cells the band resolves, in AUC.

with no band is neither resolved nor unresolved, so it can be neither kept nor dropped by the second restriction. Of those 684, 31 survive all three and the rate is 0.0%. **We therefore withdraw the claim that a one-number report misstates a sign at a rate worth quoting as a headline.** On the corpus's own AUC surface, once a sign has to be determined *and* worth acting on, disagreement is rare and the denominator is too small to carry a rate. Two corollaries follow and neither is comfortable for the conventional report. First, the reason so few cells remain is that *the data resolve so little*: 155 of the 684 AUC cells that carry a band resolve, 22.7% of them, which is the same story the whole inference surface tells at 698 of 3,420 across all 5 instruments; and 14 of the 19 pairs do not even resolve the cell a conventional paper would stand on. Second, the surface still moves, and by a distance that dwarfs the MPID: over the cells the band resolves, the mean distance from the reference increment is a median 0.0600 AUC per pair, and 0.0511 under the full restriction, both several multiples of the MPID. The one-number report is mostly not wrong about the *sign*; it is wrong about the *size*, and it does not say which.

6.5. Against specification-curve analysis as practised

The closest existing practice is specification-curve analysis (Simonsohn et al., 2020): enumerate the reasonable specifications, plot the curve, and test it against a sharp null by permutation. Supplement S17 runs it unchanged on *every* admitted pair beside the region label computed on the same cells. **They part company on 9 of the 19 pairs**, close to half: the permutation test rejects the sharp null and the same cells contain increments of *both* signs. Neither subsumes the other. The permutation test answers *is there anything here*, costs one refit per permutation per cell and needs no band; the region answers *may I quote a sign*, costs a nested bootstrap, and is what a reader deciding whether to act needs. This paper's contribution is the second, and the first should be reported beside it.

6.6. Decision rules, in sample and out

The four rules of Section 4.7 are compared under the loss defined there, inside one instrument at a time, in Table S27 and Supplement S14.3. Three results run against the surface's advocate: the conservative rule is worst in 4 instruments of 5, the one-number rule's median excess regret is *zero* in every instrument, and under rolling-origin folds the out-of-sample ordering does not hold, at 0.00885 against 0.01922. We therefore do not conclude that a surface report is a better decision rule than a number; the case rests on the decomposition and the sign disagreement, and this is a check on those.

7. Case study: a configuration management database

A configuration management database is the register an IT estate keeps of the items it contains, and it is bought partly on the argument that operational analytics need it. This section evaluates one against a named decision on a public log: will this case be reassigned to another group before it is resolved? The estate is a bank's and the log is BPI Challenge 2014 (van Dongen, 2014); the register field is the affected configuration item, which takes 2,929 distinct values on this estate's cohort at incident creation. It is reported at greater length than any of the 19 log–target pairs of Section 6 because it is the case that motivated the study and the one on which the forecast origin — when in the service-desk process the prediction is made — decides the answer.

7.1. One log, two answers, and what separates them

This log appears twice in this paper. Section 6 applies the registered generic rules of Section 5, which admit 46,606 cases and define the target by the registered handover rule. This section uses the cohort and the reassignment target defined for the estate: 45,455 cases at prevalence 0.400 (0.372 in the test half), 31,818 in training and 13,637 in test. The estate's cohort is a strict subset of the registered one: the 1,151 cases the registered rules admit on top of it are those the estate's own cutoff excludes.

The two produce different answers about the register at the later decision time, and the difference is large enough to change the sign. Against intake plus group plus the knowledge reference, the estate's cohort and reassignment target give −0.0029 and the registered cohort and handover target give +0.00768, on which the planned contrast PC3's one-sided p-value rejects at the Holm-adjusted level while its interval does not exclude zero (Section 4.4). Table S11 takes the manuscript from the first to the second one factor at a time.

It is the target, not the cohort. Over the cohort × target design the target carries 99.3% of the variance in the increment, the cohort 0.0%, and their interaction 0.7%. Reassignment to another group before resolution has prevalence 0.400; handover between activity groups, under the registered rule, has prevalence 0.927. They are not the same question and the register is not worth the same to them. Two further candidate explanations were checked and neither is one: the two analyses reach the same attribute values on every row, so it is not a data-preparation difference; and they run the same pipeline at the same reference cell, so it is not a modelling difference.

An axis nobody declares. Both analyses sort the log in time and split at seventy per cent of the row order, and 270 incidents share their timestamp with another; the sort this estate's analysis first used was not stable, so its tie order was neither declared nor reproducible, and it printed −0.0032 where the two declared rules both give −0.0029 (recomputed in the archive from that analysis's own loader, `scripts/s32_cohort.py`). Over 25 random tie-breaks the increment ranges over 0.00088 AUC without changing sign, so no conclusion turns on it — but that range is 30.6% of the increment's magnitude, produced by a choice no reporting standard asks anyone to state, which is why Supplement S21 now asks. Every number in this section is computed under a declared tie-break, by incident identifier, on the estate's cohort and reassignment target; the registered cohort's answers are among

forecast origin	baseline	n	prevalence	register levels	base AUC	with AUC	V	interval	resolved
τ_0 first touch, every call	intake	145,478	0.146	4,034	0.554	0.785	+0.2310	[+0.2204, +0.2618]	yes
τ_1 first touch, escalating only	intake	44,513	0.402	2,445	0.552	0.754	+0.2013	[+0.1930, +0.2224]	yes
τ_2 incident creation	intake	45,455	0.400	2,929	0.562	0.746	+0.1838	[+0.1725, +0.2040]	yes
τ_2 incident creation	intake + group	45,455	0.400	2,929	0.644	0.748	+0.1034	[+0.0949, +0.1295]	yes
τ_2 incident creation	intake + group + knowledge	45,455	0.400	2,929	0.805	0.802	−0.0029	[−0.0074, +0.0019]	no
τ_2 incident creation, τ_1's population	intake	44,513	0.402	2,887	0.559	0.750	+0.1914	[+0.1832, +0.2135]	yes

Table 8: The forecast origin implemented as an axis. Each moment carries its own population, its own register and its own admissible set; the *predicted target* is the same at all of them. τ_0 and τ_1 differ only in which cases are in scope. τ_1 and τ_2 differ in what is known AND in which cases are in scope, because 942 incidents have no interaction record; the sixth row is τ_2 restricted to τ_1's own population, and it is that row, not the third, that makes the τ_1-to-τ_2 step an information step alone. Every row is on the estate's own cohort and the reassignment target, under a declared tie-break, and every interval is the pointwise basic construction at 300 draws.

the 19 pairs of Section 6, so the decision-time result below is a statement about a routing decision on this estate and not about handover in general.

7.2. Three forecast origins

A forecast origin fixes three things at once — which cases are in scope, what is known about them, and which fields are admissible — and the estate's surface is estimated at each of three moments.

The register's worth turns on which one the analyst occupies: at τ_0 it is +0.2310 [+0.2204, +0.2618] over 145,478 *interactions* — calls to the desk, of which only a minority ever become incidents, which is why that population is far larger than the 45,455 incidents the later moments carry — at τ_1 +0.2013 [+0.1930, +0.2224], and at τ_2 against the intake block plus the group it is +0.1034 [+0.0949, +0.1295] while against the block, the group and the knowledge reference it is −0.0029 [−0.0074, +0.0019] — not resolvably different from zero. The step from τ_1 to τ_2 is an information step only on the matched population: 942 of the 45,455 incidents have no interaction record, so the sixth row of Table 8 restricts τ_2 to τ_1's own cases and it is that row, not the third, that isolates the information. Read as steps, the ladder moves by −0.0297 AUC from τ_0 to τ_1, which is the population; by −0.0099 from τ_1 to τ_2 on the matched population, which is the information; and by −0.0076 more from the matched population to the full one, which is the incidents without an interaction record. Supplement S14.15 gives each moment's population, register and admissible set.

7.3. The leakage tipping point, and register quality

Two further experiments on this estate are in Supplement S14.5. The first is a leakage tipping point: at τ_2 the knowledge reference takes the register's increment to -0.0029 [-0.0074, $+0.0019$], indistinguishable from zero, and writing λ for the share of that field's values assumed written *after* the incident, the increment reaches half its knowledge-free value at $\lambda = 0.64$. An unprovable admissibility assumption is therefore reported as a sensitivity analysis and not as a headline. The second is a register-quality axis under six declared mechanisms, three of them dependent, applied to both halves of the data with their parameters estimated from training alone (Supplement S11).

7.4. What this says to a configuration management programme

Supplement S18 gives the reading a service-management audience will want, stated in the estate's own vocabulary: the three forecast origins in service-desk terms, which layer of the register pays, and what would have to change for the answer to move. It is a reading of *one* estate on *one* public log and it is offered as one; Section 10 says what a second would need. Its one sentence for this section: a programme that justifies the register on a first-touch number and deploys the model at incident creation is quoting the top of a ladder and standing on the bottom of it; Section 8.3 prices that at 1.71 true positives promised per thousand cases against 0.003 delivered.

7.5. The absorption, and why the ratio is secondary

The quantity a report of this kind usually leads with is the *relative* reduction R: the share of the register's apparent value that a single free field absorbs. The absolute absorption $D = V(f \mid B_0) - V(f \mid B_1)$ is reported first, with its own interval, and R second with a Fieller set and the set's kind; on this estate at the reference cell D is $+0.0804$ [$+0.0626$, $+0.0914$] and R is 0.437 with a bounded Fieller set. The full ladder and the reason the ratio is secondary are in Supplement S19.

8. Decision-analytic evaluation

An increment in AUC is not a quantity a service desk can act on. This section reads the same surface in net benefit, which is the number of correctly nominated cases per thousand arrivals, net of the false nominations, at a declared exchange rate between the two.

8.1. The operating point is a cost ratio, and calibration comes first

At threshold θ the net benefit of a rule that nominates when $\hat{p} \geq \theta$ is

$$\mathrm{NB}(\theta) \;=\; \frac{\mathrm{TP}(\theta)}{n} \;-\; \frac{\mathrm{FP}(\theta)}{n} \cdot \frac{\theta}{1-\theta},$$

so $(1-\theta)/\theta$ is the number of false nominations the desk accepts for one true one, and choosing θ is choosing an operational cost ratio that belongs to the analyst's declaration rather than to a grid search. We report the whole grid — 31 points from 0.05 to 0.80 — and name three points a desk might occupy: $\theta = 0.10$ (9.0 false nominations per true one, a cheap review), 0.30 (2.3, a review and a miss costing about the same) and 0.60 (0.7, scarce review capacity); a reader whose desk sits elsewhere reads their own point off Figure S3. A threshold is a statement about probability, so every decision curve here is computed on *calibrated* scores, with the calibrator fitted inside the training half and rebuilt inside every bootstrap refit, and a registered rule excludes an arm whose calibration slope leaves $[0.5, 2]$; the exclusion is deliberately not repaired by recalibrating the failing arm, because recalibration is already a level of the pipeline axis and repairing an arm with it would report one specification under another's name (Supplement S14.6).

8.2. Simultaneous bands change what can be said

The decision-curve family is declared separately from the scalar surface and carries its own max-t critical value, because net benefit and a scalar instrument are not the same scale. Its critical value is the empirical quantile of Section 4.3, taken over the *admissible* family: a cell whose net-benefit difference is exactly zero in at least nine draws of ten is zero by arithmetic, not by estimation, and is excluded before the quantile is taken because it is not a statement about the register while its vanishing spread inflates the value every other cell is judged by (Section 9.2). On this estate's family that rule excludes 2 of 248 cells and gives a critical value of 6.64 [5.73, 9.08], against 7.00 with those cells kept and 3.57 from the multiplier approximation. Section 9.2 measures this estimator at 93.6% on a family matched to this one and the multiplier at 91.2%, so the band below is the better of the two and is still about a point short of nominal on the matched family — a family that reproduces the corpus's estimator ratio at 1.49 against 2.24 and so transfers less well than the surface family (Section 9.2) — which is stated here rather than in a footnote.

Across the 31 operating points of this estate's curve **the pointwise interval declares the increment resolvably positive at 24 of them and the simultaneous band at 8**, the lowest of the second at $\theta = 0.60$; the harmful counts are 0 against 0. All are built from the same draws, so what differs is the family and the level it is bought at: those points are the price of a statement about a *range* of thresholds rather than about each one separately, and a paper that reads a decision curve across its range while pricing it pointwise has claimed the first and paid for the second. The multiplier approximation would have declared 18 of the 31 resolved, at a median band width of 0.0136 against the reported band's 0.0252 and the pointwise interval's 0.0071; the order-statistic upper end of the reported critical value leaves 5, and keeping the excluded cells leaves 8, so the exclusion rule does not carry the result. Supplement S14.7 prints the whole ladder, including the measured widening of the multiplier value that an earlier construction used, and Figure S3 and Table S25 the curve. The pointwise interval is not widened by a family-wise factor; what it covers is a separate measurement, and the (n, K) calibration of Section 6.3 is not applied to this family either.

8.3. The decision, in units a service desk uses

The comparison is run in net benefit per thousand cases, over a declared distribution of exchange rates a desk plausibly holds (0.08 to 0.36, centred 0.19), and only on models that pass the registered calibration rule — which excludes 40 of 118 arms and 3 of 19 pairs. **That count is itself the strongest operational finding here**: a decision curve read off a model whose probabilities are wrong is read at a threshold the model does not have.

And the rules separate on most pairs, with the excess falling on the conservative rule rather than the one-number rule. The one-number rule's median excess is 0.000 per thousand against the conservative rule's 0.048, and the cross-pair means are 0.056 and 6.092: what the medians hide is a tail, and it is the conservative rule's. The four rules do not all lose the same amount on 11 of the 16 pairs the calibration rule leaves, and all four are identical on the other 5. Where they do separate, the one-number rule is strictly the worst on 3 and tied for worst on 2, at a maximum of 0.412 per thousand. **The case against one-number reporting does not rest on this design.** Larger than anything the choice of collapse rule costs a one-number report is the forecast origin: the register promises 1.71 per thousand at the earlier moment and delivers 0.003 at the later one, a shortfall of 1.70.

Supplement S14.10 carries the rules, the exchange-rate distribution and the exclusions; Tables S23 and S27 the numbers.

9. Simulation against a known answer

Everything so far has been measured on real logs, where the answer is not known. This section measures the estimator where it is. The full tables are in Supplement S15 and the diagnosis of the one world that failed is in Supplement S6.

9.1. What the simulation certifies of the pointwise interval

The simulation uses six worlds — four of them misspecified for the estimator — at 1,000 replicates each, with the covariate space finite so that the population AUC of any scoring rule is computed in closed form. Two estimands are separated because under misspecification they differ: V_{oracle}, the increment between the two Bayes-optimal rules, and V_{limit}, the increment between the two fitted pipelines' limits; a resampling interval can only cover the second (Supplement S6.1, Table S37). Coverage estimates carry a Monte Carlo standard error of about 0.7 percentage points, which is the resolution every comparison here should be read at (Figure S5).

Three things are established and one is not. The nested basic interval covers a median 93.0% against the fixed-model interval's 92.1%, and on the sparse world the *percentile* interval collapses to 37.2% while the basic construction restores it to 89.9%, which is why every interval and band here is the basic one (Supplement S6.2). Against V_{oracle} every construction undercovers on the drifting world — 65.8% at best — and it should, because when the coefficients move between eras the estimator is not consistent for the oracle quantity, and no resampling scheme repairs that. Varying sample size and register cardinality jointly, up to $K = 3{,}019$ levels, the ratio K/n governs the basic interval's coverage — a median 83.3% and a minimum 21.3% across the plane, short of nominal everywhere — and does not exhaust it: holding the register at one cardinality while the training half grows from 2,000 to 8,000 rows moves coverage from 0.453 to 0.613 (Supplement S9). This corpus sits in the regime where that matters, its median pair at $K/n = 0.106$ with 10 of 19 pairs above a tenth, so a nominal pointwise interval on them is narrower than it should be; past a crossing in n the basic interval stops containing its own point estimate (1.6% of replicates still straddle it by 50,000 training rows at a ratio of 0.125), which is where the widening of Supplement S9 is

undefined. The block length, varied over 3 settings around $\lceil n^{1/3} \rceil$, moves coverage by up to 3.6% within a world without changing any comparison the paper rests on.

9.2. The band's family-wise coverage

Everything above validates a *pointwise* interval. Two of this paper's three reporting objects are functions of a *simultaneous* band, and this section measures what the band's critical value covers.

The design, and why its answer is an upper bound. One draw from the sampling distribution of a synthetic family gives the point estimate, and as many further draws *from that same distribution* as the corpus family it is matched to carries give the bootstrap — so the bootstrap is not an approximation of the sampling distribution but is it. Every way a resampling scheme can be wrong — dependence it fails to reproduce, a refit that comes apart, a block length misjudged — is switched off, and what is left is the object under test: a standard error and a critical value both estimated from the same handful of draws. A coverage measured here is therefore the most the real procedure can attain, and a shortfall here is one that no improvement to the resampling repairs. The families are the ones the corpus runs — every pair's surface family of 180 cells at 400 draws, the modal decision-curve family of 1,100 cells at the same count, and the case study's own curve family of 248 cells at 200 (Table S40) — matched in cross-cell correlation, effective rank, per-cell excess kurtosis and — the statistic the verdict turns on — the ratio of the empirical quantile to the multiplier on the corpus's own families, by grid search against the corpus's draw files. The surface family reproduces that ratio at 1.25 against the corpus's median 1.50 (interquartile range 1.24 to 1.67), inside the range; the curve family reaches 1.49 against 2.24 (interquartile range 2.07 to 2.55), outside it, so its numbers below transfer less well and are read with that stated. Four regimes are run: Gaussian draws, the case the multiplier is derived for; draws carrying this corpus's tails, whose median cell has an excess kurtosis of 0.52 on the surface families and 7.9 at the ninetieth percentile; the same with the degenerate cells described below; and a fixed heterogeneous *non-zero* truth, under which a cell can be resolved correctly and family-wise coverage is the quantity a region label actually depends on. Coverage is over 2,000 replicates in each of 12 grid cells, so no coverage here is determined worse than 1.0%.

The empirical quantile attains its level on the matched family; the multiplier does not, and no draw budget makes it. On every

pair's surface family, under this corpus's tails, the empirical quantile of the observed maxima covers 94.7% against a nominal 95%, at a Monte Carlo standard error of 0.5%, and 94.6% under the non-zero truth; the order-statistic upper end of the same quantile covers 96.7%. The Gaussian multiplier covers 91.5% and 92.6% on the same families — four to six standard errors short — against 94.3% under the Gaussian draws it is derived for, where the empirical quantile covers 93.7%: the two change places where the draws are Gaussian, which is what the theory says and what the corpus's draws are not; the upper end of the multiplier's Monte Carlo interval moves that by less than a point, because it controls the error in estimating q and not the error in what q estimates, and a Rademacher wild multiplier, which keeps the observed draws' magnitudes, covers 91.4%. Along a ladder of draw counts at the surface family the multiplier rises steeply out of the smallest budgets and then *plateaus*: 80.6% at 33 draws, 90.9% at 150, 91.5% at 400 and 91.2% at 1,000. The empirical quantile behaves differently along the same ladder — 77.8%, 93.8%, 94.7% and 94.4% — because an order statistic of 33 maxima is badly determined and one of 400 is not, which is why the construction the paper reports is available at this corpus's draw count and would not have been at a third of it. The per-cell interval underneath moves 90.2% to 94.6% over the same range, so part of what any band loses it loses before a critical value is chosen. What the level costs is width: the empirical band is 1.24 times as wide as the multiplier's, and Section 6.3 prints what each resolves.

The decision-curve families, and the cells that are zero by arithmetic. At a threshold where both arms treat every case alike the net-benefit difference is zero by arithmetic, and 638 cells of the corpus's curve families are exactly zero in at least nine draws of ten, against 0 anywhere in the surface families. Standardising such a cell by its vanishing spread sends its studentised value to the largest value a deviate of B points can take, so it inflates the critical value every other cell is judged by while its own band is trivially right; the degenerate regime reproduces that, and under it the multiplier covers 92.9% on the modal curve family. A declared rule therefore removes such cells from a decision-curve family before the quantile is taken — a cell that is exactly zero in at least nine draws of ten is not a statement about the register — and Section 8.2 reports the case study's curve under the empirical quantile of the admissible family, with the excluded count printed. On the modal curve family so treated the empirical quantile covers 93.8% under the corpus's tails and 94.3% under a non-zero truth, against the multiplier's 91.3% and 92.1%; on the case study's own family, 93.6% against 91.2%. **On**

the curve families the empirical quantile is therefore about a point short of its level, on a synthetic family that under-reproduces the corpus's estimator ratio, and the article reports it as the better of the two estimators rather than as one measured at nominal; Section 8.2 says so where the counts are printed. Supplement S9.6 carries the mechanism and Table S40 every regime and candidate.

What the (n, K) calibration is, now. Supplement S9 derives a widening for the *pointwise* interval as a function of the register's cardinality over the training size, from the plane of Section 9.1. It addresses the channel this section switches off, is fitted pointwise and applied simultaneously, and is measurably misspecified in n; the article reports the labels under it as a sensitivity ladder (Supplement S9.7) and does not apply it, and Section 10 says what that leaves open.

What the simulation does not establish. It certifies the pointwise constructions on six worlds, a plane of 10 (n, K) cells and 3 block lengths, and the band's critical value on the 3 matched families of Section 9.2 under four regimes — on nothing else. The stationarity and mixing the moving-block bootstrap needs are assumed and not tested; the split point is fixed within a draw, so split-position uncertainty is outside every interval reported here; and the drifting world shows what happens when the estimator is not consistent for the quantity a reader cares about.

10. Threats to validity

Organised as threats rather than as a list. No result is computed here: every number this section quotes is the same macro, from the same result file, as the section that establishes it, so the two cannot disagree — but many are quoted twice, and a reader who wants the argument rather than the summary should read them where they are established.

Construct — this is forecast performance, not value. Nothing here measures what a register costs to acquire or maintain, whether acting on a prediction changes an outcome, or what a data programme is worth; that includes Section 7.4's layer comparison. The corpus is a family of related questions rather than replications — the two registered targets label the same case identically on a median 51% of cases, and the registers run from a configuration item to a legal article — so every corpus statistic is about *sensitivity*, every one is reported on the 8 ITSM pairs as well (Table S17), and Table S13 names

each pair's own construct threat. The registered handover target on the case study's log is close to degenerate at prevalence 0.927, carries 4,800 of 29,040 admissible cells, and is where Section 7.1 shows the register's sign turns; whether a generic handover rule is a meaningful target on ITSM logs and whether the prevalence window should close below 0.95 are open, and the rule was not narrowed after seeing which pair it admits. The quality mechanisms are applied to both halves with parameters estimated from training alone (Section 7.3), so they are the operational counterfactual and not a train–test mismatch, but they remain constructed, and validating them against a genuinely incomplete register is the obvious next study.

Internal — what the band's coverage does and does not establish. Section 9.2 measures family-wise coverage against a known answer over matched families whose bootstrap is *ideal*, so every coverage there is an upper bound on what the procedure attains. On the family matched to the design this paper reports the empirical critical value covers 94.7% under the corpus's tails and 94.6% under a non-zero truth, against a nominal 95%, at a Monte Carlo standard error of about 0.5%; the multiplier approximation covers 91.5% and 92.6%, and no draw budget closes that gap. So the region labels and ρ rest on a critical value measured at its level, with three qualifications that are limits rather than results; Supplement S20 states them, with the mechanisms this evidence retires and the further assumptions it names, one of them measured.

Conclusion — two claims a specification surface invites and this evidence does not support. Section 11 withdraws both readings a pooled surface invites, together with the sign-disagreement rate as a headline. Three qualifications are limits rather than results: both decompositions are reported under all three declared measures because there is no neutral one (the higher-order share moves by 24.3% across them), the within-instrument decomposition being primary because the decomposition inherits the metric's scale (Proposition S1); region labels are relative to an inference family of a median 16.7% of a pair's admissible cells, under a minimum resolved share that withdraws the direction on 3 of 9 pairs, and stay conditional on the quality axis being averaged over; and specification regret reports a null and is a robustness check rather than a contribution.

External — the design space is declared, not complete. $\mathcal{S}$ has nine axes in Definition 1 and the per-pair factorial varies 5 of them (Table 3); the decision time, the operating point and the cohort vary on the case study's log alone,

and the target indexes surfaces rather than varying within one. It could have more — tuning protocol, missing-data treatment, target horizon, label-noise model — so the honest claim is "the increment moves this much over *this* space". The structural omission is that prefix length, prefix bucketing and sequence encoding are not axes here and cannot be, because this study predicts at one prediction point from creation-time attributes, and those are the axes predictive process monitoring varies most; we therefore claim no result about that literature's own benchmark. A prefix axis is nonetheless run on one log (Supplement S12): holding the register and the estate fixed and moving only the prefix, the increment falls from +0.257 AUC at case creation to +0.064 two events later and +0.069 at the longest prefix run, and is undetermined in sign at 2 of 6 prefixes, so case creation — the point every other result here occupies — is the most favourable prefix for a creation-time register, and two studies reporting from the two ends of that axis would differ by a factor of 3.7 without either being wrong. One log, one register and no sequence encoder is a pilot, and the omission stands.

External — three narrower limits, named. The inference surface is smaller than the declared one and always will be, because a draw refits every arm of every cell (Section 4.2), so every region label is a claim about a fraction of the space the point estimates cover, on which *uniformly beneficial* is easier to earn. The pipeline axis is crossed on the 8 ITSM pairs and not elsewhere, and what the crossing found is not a clean replication (Section 6.2), so the case study's ordering is the common case and not a corpus fact. The case study is one bank's estate on one public log, with the availability question at its centre settled by probability rather than by a timestamped field-value history; Section 7.4 names the three changes that would move which answer. And reporting practice is not measured: the pilot that attempted it is in the archive as a hypothesis.

11. Conclusion

The incremental forecast performance of a recorded field is a functional of design choices, not a number. This paper made that operational: a declared design space, a surface over it, an exact decomposition of the surface's variance that separates the choices an analyst makes from the resampling the design also contains, simultaneous bands whose coverage is measured rather than assumed, and region labels that say where a sign is determined and withhold a direction where too little of the surface resolves to support one.

On every pair, at least one admissible specification makes the register worth less than nothing. That is a statement about point estimates and needs no band. It is also nearly guaranteed by a space that includes three-quarters-deleted registers, so the version worth carrying is the restricted one: **on 7 of 19 pairs the baseline alone — among the baselines an analyst would actually build — moves the increment further than the increment itself**. Four further results are worth carrying away.

Most of a pooled specification surface is resampling, and net of it different pairs turn on different analyst choices. The components involving the split carry a median 71.7% of the variance. Net of them the largest first-order index at the median pair is 38.9% against a higher-order share of 26.3%: the interactions carry more than any single axis on 7 of 19 pairs, against 13 before the resampling stratum is removed, so *most of the variance belongs to no single axis* is not a claim this evidence supports (Section 6.2). What it does support is that which axis leads is a property of the pair — the baseline rung on 8 pairs, the register-quality condition on 5 and the pipeline on 6. An analyst therefore cannot learn from somebody else's paper which of their own choices their answer turns on — but neither should they read a pooled specification curve as evidence that everything matters.

A one-number report is usually right about the sign and usually silent about the size. Once a sign has to be determined by the data *and* worth acting on at 0.01 AUC, the number of the 31 *AUC* cells surviving all three restrictions that contradict the conventional report is 0 — the MPID is declared in AUC and Remark 1 forbids carrying it further. **So the sign-disagreement rate is withdrawn as a headline** (Section 6.4): a sign disagreement is not a misstatement unless the sizes are worth acting on, and under the three restrictions this paper's own argument implies, they are not. What the conventional report does not say is that the cells the band *resolves* lie a median 0.0600 AUC away from the one it stands on, and that on 14 of 19 pairs the data do not determine the sign of its own cell at all.

The forecast origin is an argument of the estimand, not a preliminary. It fixes which cases are in scope, what is known about them, and which fields are admissible. On the case study, moving the prediction from first touch to incident creation moves the increment by −0.0297 AUC through the population, by a further −0.0099 through the information once the population is held fixed, and by −0.0076 more through the incidents

that have no interaction record at all — before any baseline changes; and the same log's answer changes sign when the target changes, with 99.3% of the difference on the target and 0.0% on the cohort. A desk that adopted this register on its *first-touch* number and deployed it at incident creation would be promised 1.71 true positives per thousand cases and receive 0.003.

And a threshold is not always a cost ratio. On 40 of 118 fitted models the calibration slope falls outside $[0.5, 2]$, so the threshold at which a decision curve is read cannot be interpreted as the cost ratio the decision-analytic argument requires (Section 8.1).

The practical recommendation is narrow and cheap once the surface exists: report the surface, the decomposition with the resampling stratum separated, the region and ρ with the triple they rest on; state the forecast origin and the target as parts of the estimand; declare a minimal practically important difference before reporting any disagreement rate; make the absolute increments primary; and use a ratio last, with a Fieller set, and only where its denominator is resolvably positive.

Data availability

All data are public. Event logs are obtained by DOI: BPI Challenge 2014 (https://doi.org/10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35), BPI Challenge 2013 incidents (https://doi.org/10.4121/uuid:500573e6-accc-4b0c-9576-aa5468b10cee), and the further public logs whose DOIs and SHA-256 checksums are listed in `data/corpus/CHECKSUMS.txt` — 26 files are downloaded and checksummed in all, of which the registered role-assignment rules of Section 5 admit 13, yielding the 19 log–target pairs analysed here. The counts are different quantities and Table S12 accounts for every downloaded log that is not among the admitted pairs, in one of two ways: with the registered exclusion code the rules gave it, or with the code HELD_OUT, which marks the held-out set — 1 of the checksummed files, downloaded for the out-of-corpus test and never offered to the admission rules. The literature pilot's frame, sample, codes, adjudications and evidence dossiers are in the archive; the retrieved full texts are other publishers' copyrighted PDFs and are not redistributed, but `scripts/s06_audit2.py --fetch` re-retrieves them from the DOIs in `results/s06_sample.csv`. All derived results (`results/*.csv`), all figures and all code are in the archive named in the code-availability statement.

Code availability

Code, derived results, the registered protocols, the amendment log and the verification harness are archived under a persistent identifier and developed in a public repository; both are given on the title page and are omitted here for double-blind reviewing. The archive contains a `Dockerfile` and a `requirements.lock` pinning every transitive dependency, and `REPRODUCE.md` gives the exact commands and the expected runtime.

References

Bouthillier, X., Delaunay, P., Bronzi, M., et al., 2021. Accounting for variance in machine learning benchmarks, in: Proceedings of Machine Learning and Systems (MLSys).

Breznau, N., Rinke, E.M., Wuttke, A., et al., 2022. Observing many researchers using the same data and hypothesis reveals a hidden universe of uncertainty. Proceedings of the National Academy of Sciences 119, e2203150119. doi:10.1073/pnas.2203150119.

Çarka, J., Esposito, M., Falessi, D., 2022. On effort-aware metrics for defect prediction. Empirical Software Engineering 27, 152. doi:10.1007/s10664-022-10186-7.

Chernozhukov, V., Chetverikov, D., Kato, K., 2013. Gaussian approximations and multiplier bootstrap for maxima of sums of high-dimensional random vectors. The Annals of Statistics 41, 2786–2819. doi:10.1214/13-AOS1161.

Comuzzi, M., Kim, S., Ko, J., Salamov, M., Cappiello, C., Pernici, B., 2025. On the impact of low-quality activity labels in predictive process monitoring, in: Business Process Management Workshops. doi:10.1007/978-3-031-82225-4_15.

Cook, N.R., 2007. Use and misuse of the receiver operating characteristic curve in risk prediction. Circulation 115, 928–935. doi:10.1161/CIRCULATIONAHA.106.672402.

Covert, I., Lundberg, S., Lee, S.I., 2020. Understanding global feature contributions with additive importance measures, in: Advances in Neural Information Processing Systems 33.

D'Amour, A., Heller, K., Moldovan, D., et al., 2022. Underspecification presents challenges for credibility in modern machine learning. Journal of Machine Learning Research 23, 1–61.

Davison, A.C., Hinkley, D.V., 1997. Bootstrap Methods and Their Application. Cambridge Series in Statistical and Probabilistic Mathematics, Cambridge University Press, Cambridge. doi:10.1017/CBO9780511802843.

Anonymous, 2026. Details omitted for double-blind reviewing.

van Dongen, B.F., 2014. [dataset] BPI challenge 2014. 4TU.ResearchData. doi:10.4121/uuid:c3e5d162-0cfd-4bb0-bd82-af5268819c35.

Fisher, A., Rudin, C., Dominici, F., 2019. All models are wrong, but many are useful: Learning a variable's importance by studying an entire class of prediction models simultaneously. Journal of Machine Learning Research 20, 1–81.

Ghorbani, A., Zou, J., 2019. Data Shapley: Equitable valuation of data for machine learning, in: Proceedings of the 36th International Conference on Machine Learning, pp. 2242–2251.

Holm, S., 1979. A simple sequentially rejective multiple test procedure. Scandinavian Journal of Statistics 6, 65–70.

Hutter, F., Hoos, H., Leyton-Brown, K., 2014. An efficient approach for assessing hyperparameter importance, in: Proceedings of the 31st International Conference on Machine Learning, pp. 754–762.

Jia, R., Dao, D., Wang, B., Hubis, F.A., Hynes, N., Gürel, N.M., Li, B., Zhang, C., Song, D., Spanos, C.J., 2019. Towards efficient data valuation based on the Shapley value, in: Proceedings of the 22nd International Conference on Artificial Intelligence and Statistics, pp. 1167–1176.

Kapoor, S., Narayanan, A., 2023. Leakage and the reproducibility crisis in machine-learning-based science. Patterns 4, 100804. doi:10.1016/j.patter.2023.100804.

Kaufman, S., Rosset, S., Perlich, C., Stitelman, O., 2012. Leakage in data mining: Formulation, detection, and avoidance. ACM Transactions on Knowledge Discovery from Data 6, 15:1–15:21. doi:10.1145/2382577.2382579.

Kerr, K.F., McClelland, R.L., Brown, E.R., Lumley, T., 2011. Evaluating the incremental value of new biomarkers with integrated discrimination improvement. American Journal of Epidemiology 174, 364–374. doi:10.1093/aje/kwr086.

Künsch, H.R., 1989. The jackknife and the bootstrap for general stationary observations. The Annals of Statistics 17, 1217–1241. doi:10.1214/aos/1176347265.

Liu, C., Liu, W., Guo, N., Song, R., Gu, Y., Cheng, L., 2025. Comparative evaluation of encoding techniques for workflow process remaining time prediction for cloud applications. Journal of Cloud Computing 14. doi:10.1186/s13677-025-00763-8.

Lundberg, S.M., Lee, S.I., 2017. A unified approach to interpreting model predictions, in: Advances in Neural Information Processing Systems 30, pp. 4765–4774.

Marx, C.T., Calmon, F.d.P., Ustun, B., 2020. Predictive multiplicity in classification, in: Proceedings of the 37th International Conference on Machine Learning, pp. 6765–6774.

Mohammed, S., Budach, L., Feuerpfeil, M., Ihde, N., Nathansen, A., Noack, N., Patzlaff, H., Naumann, F., Harmouch, H., 2025. The effects of data quality on machine learning performance on tabular data. Information Systems 132, 102549. doi:10.1016/j.is.2025.102549.

Patel, C.J., Burford, B., Ioannidis, J.P.A., 2015. Assessment of vibration of effects due to model specification can demonstrate the instability of observational associations. Journal of Clinical Epidemiology 68, 1046–1058. doi:10.1016/j.jclinepi.2015.05.029.

Pedregosa, F., Varoquaux, G., Gramfort, A., et al., 2011. Scikit-learn: Machine learning in Python. Journal of Machine Learning Research 12, 2825–2830.

Pencina, M.J., D'Agostino Sr., R.B., D'Agostino Jr., R.B., Vasan, R.S., 2008. Evaluating the added predictive ability of a new marker: From area under the ROC curve to reclassification and beyond. Statistics in Medicine 27, 157–172. doi:10.1002/sim.2929.

Rama-Maneiro, E., Vidal, J.C., Lama, M., 2023. Deep learning for predictive business process monitoring: Review and benchmark. IEEE Transactions on Services Computing 16, 739–756. doi:10.1109/TSC.2021.3139807.

Romano, J.P., Wolf, M., 2005. Exact and approximate stepdown methods for multiple hypothesis testing. Journal of the American Statistical Association 100, 94–108. doi:10.1198/016214504000000539.

Saltelli, A., Annoni, P., Azzini, I., Campolongo, F., Ratto, M., Tarantola, S., 2010. Variance based sensitivity analysis of model output. Computer Physics Communications 181, 259–270. doi:10.1016/j.cpc.2009.09.018.

Semenova, L., Rudin, C., Parr, R., 2022. On the existence of simpler machine learning models, in: Proceedings of the 2022 ACM Conference on Fairness, Accountability, and Transparency, pp. 1827–1858. doi:10.1145/3531146.3533232.

Senderovich, A., Di Francescomarino, C., Ghidini, C., Jorbina, K., Maggi, F.M., 2018. Intra and inter-case features in predictive process monitoring. Information Systems 74, 102–113. doi:10.1016/j.is.2017.10.005.

Silberzahn, R., Uhlmann, E.L., Martin, D.P., et al., 2018. Many analysts, one data set: Making transparent how variations in analytic choices affect results. Advances in Methods and Practices in Psychological Science 1, 337–356. doi:10.1177/2515245917747646.

Simonsohn, U., Simmons, J.P., Nelson, L.D., 2020. Specification curve analysis. Nature Human Behaviour 4, 1208–1214. doi:10.1038/s41562-020-0912-z.

Sobol', I.M., 2001. Global sensitivity indices for nonlinear mathematical models and their Monte Carlo estimates. Mathematics and Computers in Simulation 55, 271–280. doi:10.1016/S0378-4754(00)00270-6.

Steegen, S., Tuerlinckx, F., Gelman, A., Vanpaemel, W., 2016. Increasing transparency through a multiverse analysis. Perspectives on Psychological Science 11, 702–712. doi:10.1177/1745691616658637.

Strobl, C., Boulesteix, A.L., Kneib, T., Augustin, T., Zeileis, A., 2008. Conditional variable importance for random forests. BMC Bioinformatics 9, 307. doi:10.1186/1471-2105-9-307.

Suriadi, S., Andrews, R., ter Hofstede, A.H.M., Wynn, M.T., 2017. Event log imperfection patterns for process mining. Information Systems 64, 132–150. doi:10.1016/j.is.2016.07.011.

Teinemaa, I., Dumas, M., La Rosa, M., Maggi, F.M., 2019. Outcome-oriented predictive process monitoring: Review and benchmark. ACM Transactions on Knowledge Discovery from Data 13, 17:1–17:57. doi:10.1145/3301300.

VanderWeele, T.J., Ding, P., 2017. Sensitivity analysis in observational research: Introducing the e-value. Annals of Internal Medicine 167, 268–274. doi:10.7326/M16-2607.

Verenich, I., Dumas, M., La Rosa, M., Maggi, F.M., Teinemaa, I., 2019. Survey and cross-benchmark comparison of remaining time prediction methods in business process monitoring. ACM Transactions on Intelligent Systems and Technology 10, 34:1–34:34. doi:10.1145/3331449.

Vickers, A.J., 2008. Decision analysis for the evaluation of diagnostic tests, prediction models, and molecular markers. The American Statistician 62, 314–320. doi:10.1198/000313008X370302.

Vickers, A.J., van Calster, B., Steyerberg, E.W., 2019. A simple, step-by-step guide to interpreting decision curve analysis. Diagnostic and Prognostic Research 3, 18. doi:10.1186/s41512-019-0064-7.

Vickers, A.J., Elkin, E.B., 2006. Decision curve analysis: A novel method for evaluating prediction models. Medical Decision Making 26, 565–574. doi:10.1177/0272989X06295361.

Vickers, A.J., Van Calster, B., Steyerberg, E.W., 2016. Net benefit approaches to the evaluation of prediction models, molecular markers, and diagnostic tests. BMJ 352, i6. doi:10.1136/bmj.i6.

Weytjens, H., De Weerdt, J., 2022. Creating unbiased public benchmark datasets with data leakage prevention for predictive process monitoring, in: Business Process Management Workshops, pp. 18–29. doi:10.1007/978-3-030-94343-1_2.

Williamson, B.D., Gilbert, P.B., Carone, M., Simon, N., 2021. Nonparametric variable importance assessment using machine learning techniques. Biometrics 77, 9–22. doi:10.1111/biom.13392.